

Final Credit Model Project

Credit Models

José Armando Melchor Soto – 745697
Rolando Fortanell Canedo – 744872
David Campos Ambriz – 7444435

14/05/2004

Table of Contents

Executive Summary	2
1. Introduction	4
1.1 Description of the chosen credit model.	4
1.2 Justification of its relevance in the financial industry.	5
1.2.1 Economic Magnitude of Remittance Flows to Mexico and Jalisco	5
1.2.2 Exclusion Despite Stable Income	6
1.2.3 Destination, Productivity, and the Opportunity of Remittances	7
2. Model Development: Modeling Framework	8
2.1 Data & Methodology	8
2.1.1 Data Description	8
2.1.2 Data Modeling	10
2.1.3 Description Variables	17
2.1.4 Class Distribution of the Target Variable	18
2.1.5 Variable Type	19
2.1.7 Preprocessing	21
2.2 Exploratory Feature Analysis	22
2.3 Feature Importance & Explainability	26
2.3.1 XGBoost Hyperparameter Optimization	28
2.4 SHAP Analysis	28
2.5 Feature Selection	30
3. Model Architecture & System Design	34
3.1 Architecture	34
3.1.1 Project Structure	34
3.1.2 Object-Oriented Design	34
3.1.3 Credit Risk System Architecture	35
3.2 System Design	35
3.2.1 ML System Pipeline Overview	35
3.2.2 Model Evaluation	39
3.3 Business-Oriented Model Comparison	42
3.3.1 Hyperparameter Optimization Framework	43
3.4 Experimental Results	44
3.5 Performance Evaluation Results	46
3.5.1 ROC - AUC	47
3.5.2 Confusion Matrix	48
3.5.3 Predicted PD Distribution	49
3.5.4 Real vs Predicted PD Distributions by Class	50
3.5.5 Risk Bucket Analysis	51
3.6 Final Model Selection	52
3.6.1 Final Hyperparameter Configuration	53



3.7	Credit rate construction	53
3.8	Model Validation.....	57
3.8.1	Validation techniques and results obtained	57
3.8.2	Analysis of the model's accuracy and effectiveness.....	58
3.8.3	Comparison with the unsecured personal loan segment.....	59
3.8.4	Financial Model: Portfolio performance and model performance	60
3.9	Practical Application	61
3.9.1	Specific industry use case.	61
3.10	Simulation and results analysis.	62
3.10.1	Financial projections for 3 years.....	62
3.11	Impact on existing processes and problem-solving.....	64
3.12	Deployment and ML System Integration	65
3.12.1	Interactive Geospatial Dashboard.....	66
3.12.2	Machine Learning Monitoring and Analytics Dashboard	67
4.	Impact Assessment	68
5.	Triple impact assessment (social, environmental, and financial).....	69
6.	Business Model	70
6.1	Data Construction	71
6.2	Credit Rate Construction.....	72
7.	Model Limitations & Assumptions & Future Improvements	73
8.	Final Discussion & Conclusions.....	75
9.	References	78
10.	Appendices	82



Executive Summary

Mexico received USD 64.745 billion in remittances during 2024, a record high after eleven years of growth. During this period, Jalisco has positioned itself as the third largest recipient state, with USD 5.503 billion, equivalent to 8.5% of the national total (Bank of Mexico, 2025; IIEG, 2025). However, despite the magnitude and regularity of these flows, the households that receive them remain excluded from formal credit markets. Almost 90% of remittances arrive via electronic transfer and are deposited directly into the recipient's bank account (BBVA, 2025), generating a verifiable, recurring, and traceable income record that the credit system still ignores. As a result, millions of households in rural and semi-urban municipalities of Jalisco receive international income each month, yet they are ultimately unable to access housing financing due to a lack of payroll documentation or a credit history.

This project develops a comprehensive credit scoring system designed to address this problem by creating a home improvement loan or mortgage product for remittance-receiving households in Jalisco, with loan amounts between MXN 50,000 and 500,000, terms of 5 to 15 years, and payments via automatic debit from the same account where the household receives its remittances. The main proposition is that the temporal pattern of remittance flows, observed over a 24-month period, can serve as a substitute for credit history to predict repayment capacity and the probability of default.

The dataset supporting the model consists of 10,000 synthetic household records, each calibrated with official sources such as Banxico, CONAPO, INEGI, CNBV, CEMLA, BBVA Research, and IIEG Jalisco. Because individual remittance transaction data are not publicly available, a two-stage data construction process was implemented. In the first stage, quarterly municipal remittance data from Banxico's CE166 series were disaggregated to monthly frequency using the Chow-Lin temporal disaggregation method, with the national monthly series (SE27803) as the indicator variable. From each simulated history, 27 predictor variables were constructed, organized into three analytical pillars: Pillar A captures the temporal pattern of the remittance flow (12 characteristics, including the remittance-quota ratio, the coefficient of variation, the months without transfers, and the slope of the trend); Pillar B captures the sociodemographic characteristics of the households (6 characteristics,

including age, education, gender, and housing tenure); Pillar C captures the municipal socioeconomic and migration context (9 characteristics, including the Migration Intensity Index, the Marginalization Index, and a composite Regional Score calculated for the 124 municipalities of Jalisco).

The target variable, `default_12m`, reflects a default rate of 5.5%, calibrated against CNBV mortgage portfolio delinquency data and adjusted upward to reflect the higher-risk profile of the target segment. The class imbalance present in the dataset (94.5% non-defaulted, 5.5% defaulted) was addressed through stratified sampling and the use of assessment metrics robust to imbalance, including ROC-AUC. For the evaluation using machine learning models, four machine learning algorithms were evaluated (Logistic Regression, Random Forest, XGBoost, and LightGBM), and hyperparameter optimization was performed with Optuna.

The product's financial viability was assessed using a three-year projection model built for a non-bank financial institution (ENR SOFOM), originating 500 loans in Year 1, increasing to 650 in Year 2 and 845 in Year 3, with an average loan size of MXN 200,000. The model projects a gross loan portfolio of MXN 320.9 million for Year 3, with net income increasing from MXN 558,888 in Year 1 to MXN 29.8 million in Year 3, a capital recovery period of 1.67 years, and a capital adequacy ratio (CAPR) consistently above the Basel III minimum of 10.5%. All 13 financial validations of the model are met, including a positive net present value and an internal rate of return (IRR) on equity exceeding the cost of capital

As will be seen in the product developed below, this project goes beyond financial performance. By making remittance flows recognizable as evidence of creditworthiness, the model allows approximately 4.9 million households nationwide, and hundreds of thousands in Jalisco specifically, to access formal housing credit for the first time (Cervantes and Ostolaza, 2022; CEMLA, 2025).

1. Introduction

1.1 Description of the chosen credit model.

Throughout this project, a mortgage loan model for home improvements will be developed. This product is designed to target Mexican households that receive remittances, with a special focus on the state of Jalisco.

The proposal itself consists of a medium- to long-term credit product, with amounts between 50,000 and 500,000 pesos and terms of 5 to 15 years. In this case, the proposed evaluation is not based on assessing repayment capacity through formal employment income, but rather on the observed pattern of remittance received over a period of at least 24 months. This proposed analysis will also be complemented with sociodemographic variables of the recipient and contextual variables of the municipality in which the interested person resides.

Unlike traditional mortgage products, where risk assessment is based on the borrower's credit history and salary or payroll, the proposed model is based on the recognition that remittances received periodically can represent, in practice, a documentable, predictable, and verifiable income stream through the existing banking system. According to recent records, 99% of remittances arriving in Mexico in 2024 were made via electronic transfers (Bank of Mexico, 2025), meaning that each transfer from abroad leaves a record and can be reconstructed as an individual time series for credit assessment purposes, just as one would analyze a person's monthly salary.

The proposed model is built around three main components. The first is a module for acquiring and constructing variables. It takes the monthly series of remittances received by the household over the last 24 to 36 months and, based on this, compiles a set of descriptive statistics on the flow's behavior, including the monthly median, variance, number of months without remittances, trend, seasonal decomposition, age of the flow, and sender concentration. The second module takes and combines these temporal variables with traditional variables (age, number of dependents, education level, type of housing) and with contextual variables of the municipality (migration intensity index, proportion of receiving households). The third module analyzes the characteristics to construct a score and

classification, and based on this, creates an interpretable risk scale, allowing for the assignment of differentiated rates, maximum amounts, and terms.

This financial product is designed to offer two options tailored to families. The first is a mortgage loan to purchase a home (new or used) in their municipality of residence; in this case, a family member or person living abroad can function as a guarantor or manage the payments directly. The second option is a loan to expand, remodel, or build on their own land, a common practice among families receiving remittances. In fact, this model has already been successfully implemented in previous CONAVI programs (2017). To simplify the process, payments can be automatically deducted from the same account where the money is received, making it much safer and easier for everyone.

1.2 Justification of its relevance in the financial industry.

1.2.1 Economic Magnitude of Remittance Flows to Mexico and Jalisco

Remittances are currently one of the most important sources of foreign currency for the Mexican economy and, in terms of households, represent a financial transfer with a frequency and magnitude comparable to programmable spending on social programs. According to the Bank of Mexico, Mexico received \$64.745 billion (USD) in remittances during 2024, a record high after 11 years of continuous growth, although these flows have recently been limited (2.3% nominal annual growth compared to the double-digit rates observed between 2020 and 2022) (Bank of Mexico, 2025). Likewise, 13.7 million remittance transactions were recorded during 2024, with an average remittance of \$393 USD (BBVA Research, 2025). And, by 2025, the first annual contraction in remittance flows since 2013 was observed, ending at \$61.791 billion (USD) (a 4.6% decrease compared to 2024). This decrease is mainly attributed to the tightening of US immigration policy, increased scrutiny of financial transfers, and the proposed tax on remittances currently under debate in the United States (Líder Empresarial, 2026).

Regarding the overall context in Mexico, Jalisco remains one of the most important states for this flow. According to the Jalisco Institute of Statistical and Geographic Information (IIEG, 2025), the state was the third largest recipient of remittances in Mexico during the fourth

quarter of 2024, capturing 8.6% of the national total. By the end of 2024, households in Jalisco had received \$5.503 billion in remittances, representing approximately 8.5% of the national total (Bank of Mexico, 2025). Similarly, in 2025, Jalisco maintained its third-place ranking despite the overall economic downturn, receiving \$5.145 billion, equivalent to 8.3% of the national flow, although with a decrease of almost 10%, greater than the national average (Líder Empresarial, 2026). Jalisco's greater sensitivity to the impact of migration is explained by the concentration of its recipient households in rural and semi-urban areas, where remittances represent a significant source of income for some families.

1.2.2 Exclusion Despite Stable Income

Despite the magnitude and regularity of the flow, households receiving remittances in Mexico face lower financial inclusion than the general population, preventing them from using this income stream to acquire assets that would help them obtain credit. According to measurements by the Center for Latin American Monetary Studies (CEMLA), based on microdata from the 2021 National Survey of Financial Inclusion, approximately 11.1 million people and 4.9 million households in Mexico receive remittances (Cervantes and Ostolaza, 2022). 12.3% of the country's adults are remittance recipients, with a higher proportion among women (13.6%) than among men (10.9%). The Bajío Occidental region, which includes Jalisco, has the highest proportion of remittance-receiving households in the country, with one in every 3.1 households in localities with fewer than 2,500 inhabitants receiving this income from abroad (CEMLA, 2025).

Although nearly 90% of remittances to Mexico are now deposited into the recipient's account (BBVA, 2025), receiving remittances in an account is not commonly considered by the credit system. Since a person's account functions as a cash withdrawal point, not as a record of repayment capacity for loan approval, this disconnect means that millions of households with documented and predictable incomes are excluded from access to formal mortgages and have no option but to resort to informal financing, spending their own savings, or, in the best-case scenario, government subsidy programs.

1.2.3 Destination, Productivity, and the Opportunity of Remittances

The use of remittances in Mexico demonstrates the need for more accessible housing loans. Currently, this money is primarily used to cover basic needs, and 72% of these households are headed by women (BBVA, 2025). However, many families build their homes gradually: saving for months to buy materials and proceed in stages. This method is inefficient, as material prices rise over time, and the family has to live in an unfinished house for years. A loan that takes future remittances into account would allow families to receive the money now to finish their homes at once, paying it off later with the same support they already receive from abroad.

2. Model Development: Modeling Framework

2.1 Data & Methodology

2.1.1 Data Description

The dataset used in this project was constructed from official public sources, since individual transactional information on remittance flows per household is not publicly available due to confidentiality reasons. This limitation, common in credit analysis of informal segments (World Bank Group, 2019), was the reason behind the construction of a high-fidelity synthetic dataset, in which each parametric distribution was calibrated with empirical evidence from ten official institutions and studies.

The sources used can be grouped into four functional categories:

The Bank of Mexico.

Which contributed two fundamental series: the monthly national series of family remittances (code SE27803) from the Economic Information System (SIE), with 144 observations from January 2013 to December 2024, and the quarterly remittance data by municipality for Jalisco (table CE166), covering all 124 municipalities in the state with 48 quarters of historical data. The accumulated national total for 2024 for series SE27803 amounts to USD 64,746 million, replicating, with a difference of USD 1.38 million, the official figure of USD 64,745 million reported by BBVA Research (2025), thus validating the integrity of the download process. The municipal series, in turn, replicates the official Jalisco figure for 2024 (USD ~5.503 billion), with a difference of USD 20.74 million (Banxico, 2025).

CONAPO

Which provide the 2020 Municipal Migration Intensity Index (DP2 methodology) and the 2020 Municipal Marginalization Index, which were incorporated as variables in Pillar C (municipal context) of the model.

- The INEGI Population and Housing Census 2020

Contributed the municipal ITER indicators for housing, education, and composition of the employed population, used both in Pillar C and to calibrate the demographic distributions of the synthetic register.

The CNBV (2024)

Who provided delinquency rate data by portfolio type, used in calibrating the target variable for twelve-month default (default_12m). The IIEG Jalisco, BBVA Research (2025) and CEMLA completed the table of sources with reports of migratory intensity in Jalisco, the Mexico Migration and Remittances Yearbook 2024, and the demographic characteristics of households receiving remittances, respectively (see Table I).

Data Sources and Main Uses		
Source	Data	Main use
Banxico, SIE	Monthly domestic remittances (SE27803)	Component 1, Chow-Lin indicator
Banxico, SIE	Quarterly remittances by municipality (CE166)	Regional Score Target
Banxico, Inf. Operativa	Housing portfolio and mortgage delinquency	Default rate calibration
CONAPO	Migration Intensity Index 2020 (DP2)	Pillar C, Migration Context
CONAPO	Municipal Marginalization Index 2020	Pillar C, socioeconomic context
INEGI, 2020 census	Municipal ITER Indicators	Pillar C, characteristics of the municipality
CNBV	Delinquency rate by portfolio type	Target default rate calibration
IIEG Jalisco	Remittances and Migration Intensity	Cross-validation of State Figures
BBVA Research	2024 Migration and Remittances Yearbook	Calibration of census distributions
CEMLA (Center for Latin American Monetary Studies)	Demographic characteristics of recipient households	Calibration of census distributions

Table 1 : Monthly seasonality of remittances: national series (SE27803) vs. Jalisco disaggregated by Chow-Lin.

The final dataset contains 10,000 records, one per hypothetical household, with 30 variables distributed across the binary target variable (default_12m) and 27 predictor variables organized into three groups:

Pillar A — Temporal pattern of remittance flow (12 features): calculated over a 24-month window (January 2023 and December 2024) simulated individually for each household. These include the median and mean remittance amount in months with remittances, standard deviation, coefficient of variation, normalized trend slope, seasonal strength, total number and maximum streak of months without remittances, age of the flow in months, proposed hypothetical loan payment in Mexican pesos, and the remittance-payment ratio.

Pillar B — Demographic characteristics of the recipient (6 features): age, female gender indicator, number of dependents, education level (ordinal variable 0 to 4), and binary indicators of homeownership and renting.

Pillar C — Contextual variables of the municipality (9 features): Migration Intensity Index DP2, percentage of households receiving remittances and households with migrants, Municipal Marginalization Index, average municipal schooling, percentage of employed population with income up to two minimum wages, and the three variables of the Regional Score developed in Component 1 of the project (total score, sub-score for ability to pay, and sub-score for migration intensity).

The target variable `default_12m` is binary and indicates whether the household defaulted within twelve months of the observation cutoff. The target default rate was calculated with reference to the mortgage portfolio delinquency rate reported by the CNBV (2024) for the housing segment, adjusted upwards to reflect the higher risk profile of the target segment, i.e., households without a formal credit history and without explicit mortgage collateral.

2.1.2 Data Modeling

Temporal disaggregation: Chow-Lin method

The first step in generating this dataset was to resolve the frequency inconsistency between the publicly available remittance series. The Bank of Mexico publishes the national series (SE27803) on a monthly basis and the municipal series for Jalisco (Table CE166) on a quarterly basis. Since Pillar A of the model requires monthly resolution to construct the features of the individual flow pattern, the temporal disaggregation procedure of Chow and Lin (1971) was used, whose mathematical formalization is as follows:

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad u_t \sim AR(1)$$

where y_t represents the unobserved monthly series of the municipality, x_t is the observable national indicator series on a monthly frequency basis, and u_t is a first-order autoregressive process that captures the residual dynamics not explained by the indicator variable. This method, used by statistical institutes such as Eurostat, the International Monetary Fund, and INEGI to produce high-frequency estimates from low-frequency data, guarantees that the sum of the three estimated months of each quarter exactly reproduces the observed quarterly value (Eurostat, 2003; Vera-Jaramillo, 2025; Mosley, Eckley, and Gibberd, 2022).

The project implementation performed a grid search on the autoregressive parameter ρ between 0 and 0.99 with increments of 0.01, selecting for each municipality the value that minimizes the generalized sum of squares of the residuals. The result was a new file, "Jalisco_municipal_remesas_mensuales.csv", with 17,856 observations corresponding to 124 municipalities over 144 months. Validation of the aggregation constraint confirmed that the maximum absolute error is on the order of 10^{-10} million dollars, that is, practically zero. As further validation, the seasonal factors of the disaggregated series for Jalisco reproduce the pattern of the national indicator series with a correlation of $r = 0.9842$ (Figure I), with a seasonal peak in December (factor 1.0669, corresponding to the Christmas bonus) and a trough in January (0.8496), which confirms the intra-annual consistency of the procedure.

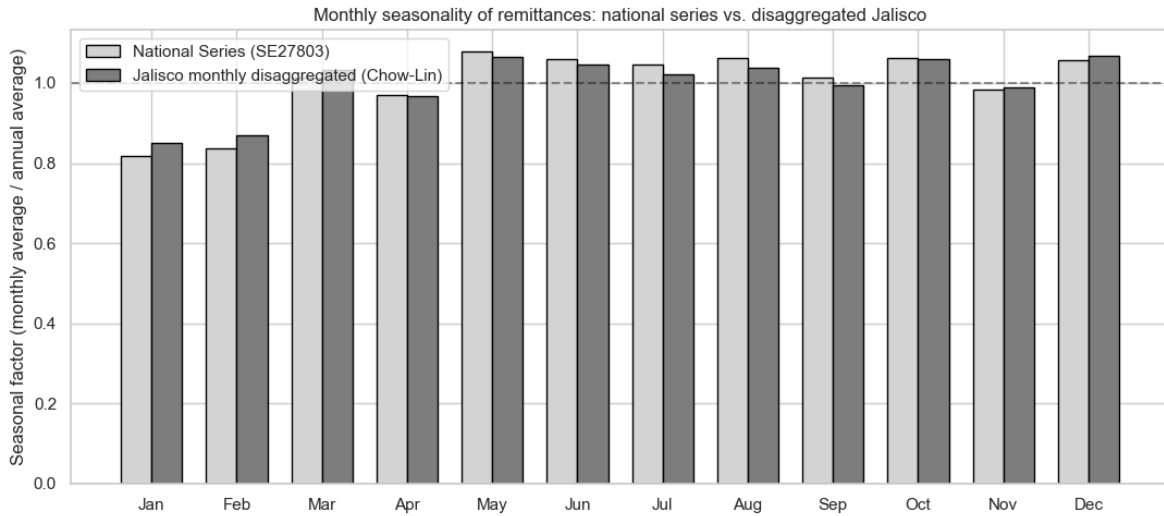


Figure II : Monthly seasonality of remittances: national series (SE27803) vs. Jalisco disaggregated by Chow-Lin.

Contextual variables and Regional Score

For Component 1 of the model, the Regional Score was constructed as a metric on a [0,1] scale that combines the municipality's ability to pay (based on average years of schooling, percentage of the population with incomes above two minimum wages, and a proxy for access to housing) and its exposure to the risk of migration flows (migration intensity DP2, marginalization, and remittance dependency as a percentage of households). The file `jalisco_municipal_features.csv` contains the 37 base variables for the 124 municipalities, on which the score was calculated.

The distribution of the Regional Score shows a median of 0.4852 (P25-P75: 0.4495-0.5212), with the municipalities in the Guadalajara metropolitan area and those with diversified economies (Puerto Vallarta, Tlajomulco) in the highest positions, and the municipalities with the greatest marginalization and remittance dependency in the lowest (Figure III).

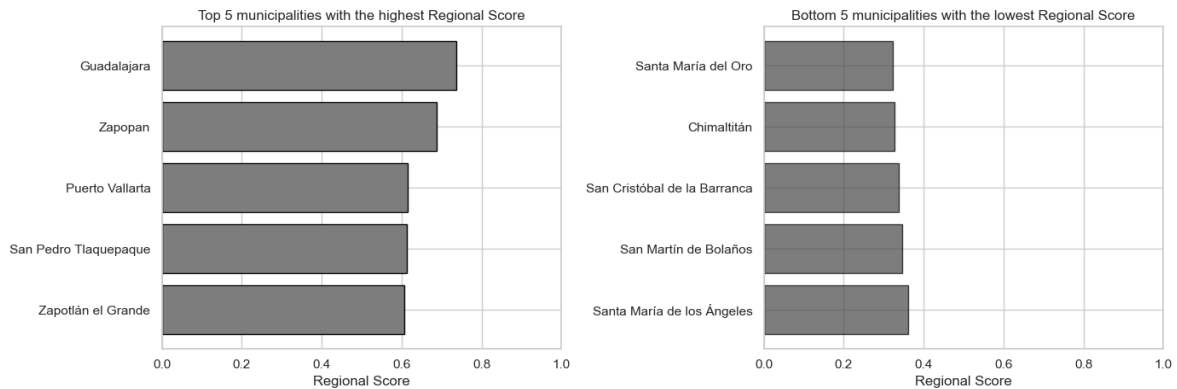


Figure III : Top 5 and Bottom 5 municipalities of Jalisco by Regional Score.

Generation of the synthetic household registry

With the previous step completed, the next thing to do was to construct an individual registry of 10,000 synthetic households whose characteristics were calibrated against distributions reported by official institutions. The households were geographically distributed proportionally to the actual volume of remittances reported for each municipality in 2024, resulting in an allocation correlation of $r = 1.0000$ with the actual figures (Banxico, 2025).

The most represented municipalities are Guadalajara (1,203 households, 12.0%) and Zapopan (712 households, 7.1%).

The demographic distributions adopted for each variable in the registry are as follows, calibrated against official sources:

- Age : limited in a normal distribution in the interval $[18, 80]$, with a mean of 45 years and a standard deviation of 12, aligned with the range reported by CEMLA and BBVA Research (2024) for remittance recipients in Mexico.
- Gender : probability of 0.65 for female, consistent with the fact that between 60% and 70% of remittance recipients in Mexican households are women (BBVA Research, 2025).
- Education level: a multinomial distribution with probabilities $[0.08, 0.32, 0.28, 0.20, 0.12]$ for the levels of no schooling, primary, secondary, upper secondary, and higher education, calibrated with microdata from the 2020 INEGI Census for recipient households in high-migration states in the central-western region of the country.
- Number of dependents: Poisson distribution on the interval $[0, 8]$ with parameter $\lambda = 2.3$, consistent with the average size of the receiving household (≈ 3.3 people) reported by CEMLA.
- Current housing type: categorical distribution $[0.62, 0.22, 0.16]$ for owned, rented, and borrowed, calibrated with tenure data for households receiving remittances in Jalisco according to the 2020 INEGI Census.
- Duration of remittance flow: consists in a lognormal distribution on $[24, 240]$ months with $\mu = \ln(60)$ and $\sigma = 0.70$, equivalent to a median of 60 months, in line with the average duration of the flow reported in studies on Mexican migration dynamics (BBVA Research, 2025).
- Expected median monthly remittance: made as a lognormal distribution with $\mu = \ln(380)$ and $\sigma = 0.45$, calibrated with the average amount per shipment of USD 393 reported by Banxico for 2024, with a realized median of USD 376 (within 1% of the target) (BBVA Research, 2025).

The validation of the distributions confirmed that the average age (45.2 years) coincides with the CEMLA calibration, the proportions of schooling and housing tenure replicate almost exactly the percentages of the INEGI 2020 Census, and the median monthly remittance is within 1% of the Banxico target (Figure IV).

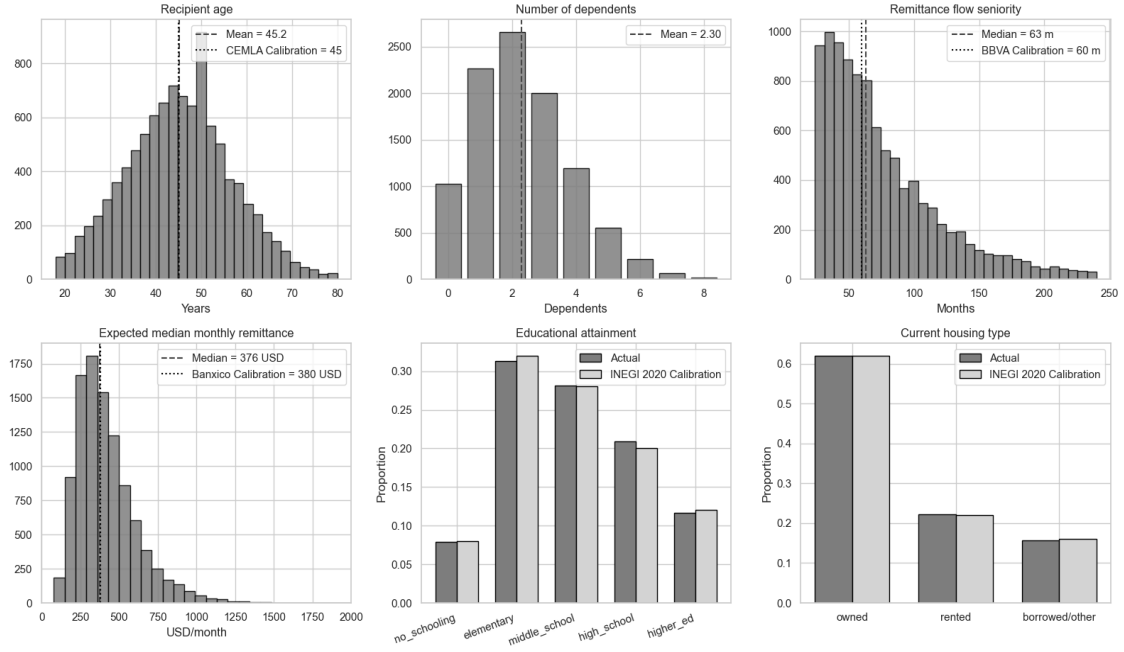


Figure IV : Marginal distributions of Pillar A and demographic variables of the synthetic register: comparison between realized values and target calibrations.

Simulation of individual monthly series

For each of the 10,000 households, a simulated 24-month transactional history was generated, which constitutes the input for Pillar A. Each monthly series was constructed using a multiplicative model with three components:

$$remittances_{i,t} = M_i \cdot s(month_t) \cdot \varepsilon_{i,t} \cdot 1_{transfer_{i,t}}$$

where M_i is the monthly base amount of the household (taken from the synthetic record), $s(month_t)$ is the seasonal factor for Jalisco estimated in Component 1, $\varepsilon_{i,t}$ is a lognormal shock with a unit mean and a coefficient of variation of 0.30 (calibrated with the residual monthly dispersion of the 2013–2024 national series), and $1_{transfer_{i,t}}$ is a binary indicator with a probability of 0.95 that captures the possibility of months without remittances,

consistent with the 5% of receiving households that experience at least one month without remittances per year according to BBVA Research (2025).

The percentage of months without shipments was 4.95%, within the theoretical calibration range (5%). The correlation between the normalized seasonal pattern of the aggregated synthetic record and the actual series for Jalisco 2023–2024 was $r = 0.8055$ (see Figure 5), confirming that the dataset adequately captures the actual seasonal dynamics of the market.

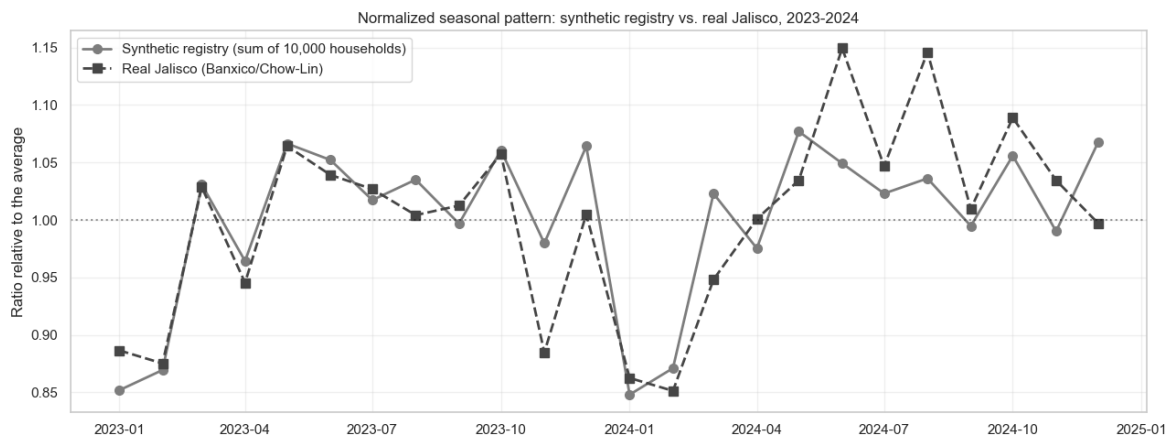


Figure V : Normalized seasonal pattern: synthetic record (sum of 10,000 households) vs. real Jalisco

The distribution of interruptions per household (number of months without shipment in the 24-month window) follows a Binomial(24, 0.05) distribution with a realized mean of 1.19 and variance of 1.14, both in line with the expected theoretical parameters.

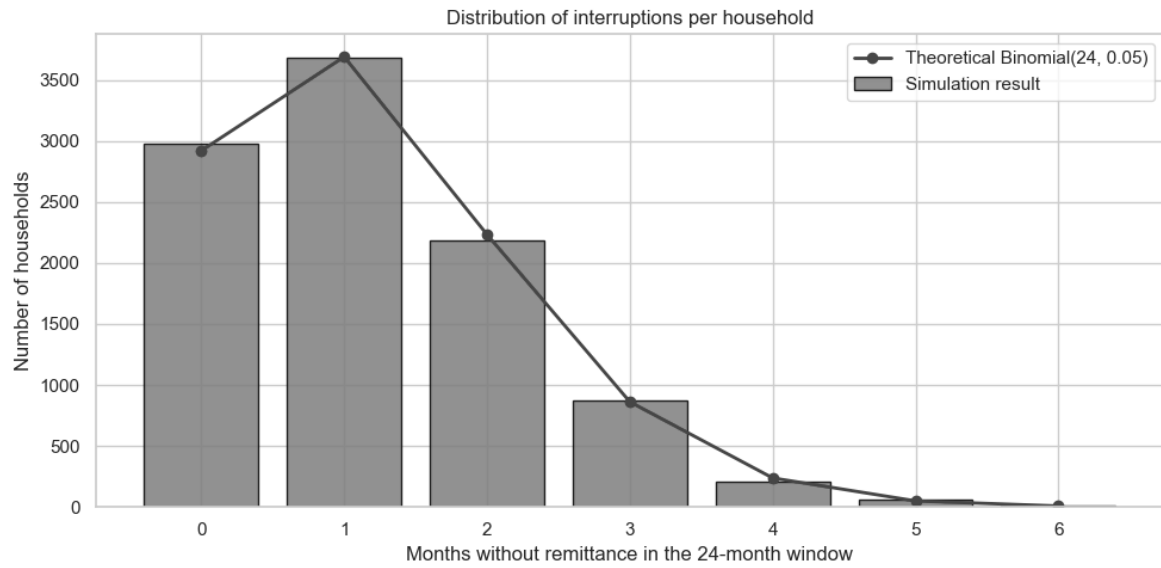


Figure VI : Distribution of interruptions per household in the 24-month window, vs. theoretical Binomial distribution(24, 0.05)

2.1.3 Description Variables

Table II presents the complete variable dictionary of the remittance-based credit risk dataset. Identification variables such as *id_hogar*, *cve_municipio*, and *municipio* were retained exclusively for traceability and regional analysis purposes; however, they were excluded from the predictive modeling stage, as they do not provide direct predictive value and could introduce non-generalizable patterns or potential leakage into the model.

Variable	Type	Description	Range / Values
<i>id_hogar</i>	Integer	Unique household identifier	Positive integer
<i>cve_municipio</i>	Integer	Municipality code	Official municipality codes
<i>municipio</i>	Categorical	Municipality name	Text
<i>pa_mediana_remesa</i>	Float	Median remittance amount	MXN ≥ 0
<i>pa_media_remesa</i>	Float	Average remittance amount	MXN ≥ 0
<i>pa_desv_remesa</i>	Float	Remittance variability	MXN ≥ 0
<i>pa_cv_remesa</i>	Float	Relative remittance volatility	≥ 0
<i>pa_pendiente_24m</i>	Float	24-month remittance trend	Negative to positive
<i>pa_pendiente_norm</i>	Float	Normalized remittance trend	Standardized values
<i>pa_n_meses_sin_envio</i>	Integer	Months without remittances	0 – 24
<i>pa_max_racha_sin_envio</i>	Integer	Longest consecutive gap without remittances	≥ 0
<i>pa_fuerza_estacional</i>	Float	Seasonal remittance strength	0 – 1
<i>pa_antiguedad_meses</i>	Integer	History length in months	≥ 0
<i>pa_cuota_propuesta_mxn</i>	Float	Proposed loan installment	MXN ≥ 0
<i>pa_ratio_remesa_cuota</i>	Float	Remittance-to-installment ratio	≥ 0
<i>pb_edad</i>	Integer	Beneficiary age	Years
<i>pb_genero_F</i>	Binary	Female beneficiary indicator	0 = No, 1 = Yes
<i>pb_n_dependientes</i>	Integer	Number of dependents	≥ 0
<i>pb_escolaridad_ord</i>	Ordinal	Education level	Ordered categories
<i>pb_vivienda_propia</i>	Binary	Owns the house	0 = No, 1 = Yes
<i>pb_vivienda_rentada</i>	Binary	Rents the house	0 = No, 1 = Yes
<i>pc_iim_dp2</i>	Float	Migration intensity index	Continuous
<i>pc_pct_viv_remesas</i>	Float	% of homes receiving remittances	0 – 100
<i>pc_pct_viv_emigrantes</i>	Float	% of homes with emigrants	0 – 100
<i>pc_imn_2020</i>	Float	Socioeconomic marginalization index	Continuous
<i>pc_escolaridad_promedio_mun</i>	Float	Average municipal education	Years
<i>pc_pct_pob_ocupada_2sm</i>	Float	% earning up to two minimum wages	0 – 100
<i>pc_score_regional</i>	Float	Regional socioeconomic score	Continuous
<i>pc_sub_capacidad_pago</i>	Float	Regional payment capacity subscore	Continuous
<i>pc_sub_intensidad_migratoria</i>	Float	Regional migration subscore	Continuous
<i>default_12m</i>	Binary	Default within 12 months	0 = No default, 1 = Default

Table II : Description of Model Variables and Data Types

2.1.4 Class Distribution of the Target Variable

Figure VII summarizes the distribution of the target variable (default_12m) in the training dataset. The data exhibits a significant class imbalance, where the Non-Default class represents approximately 94.5% of the observations (9,446 records), while the Default class accounts for only 5.5% (554 records). This indicates a strong dominance of non-default observations over default cases.

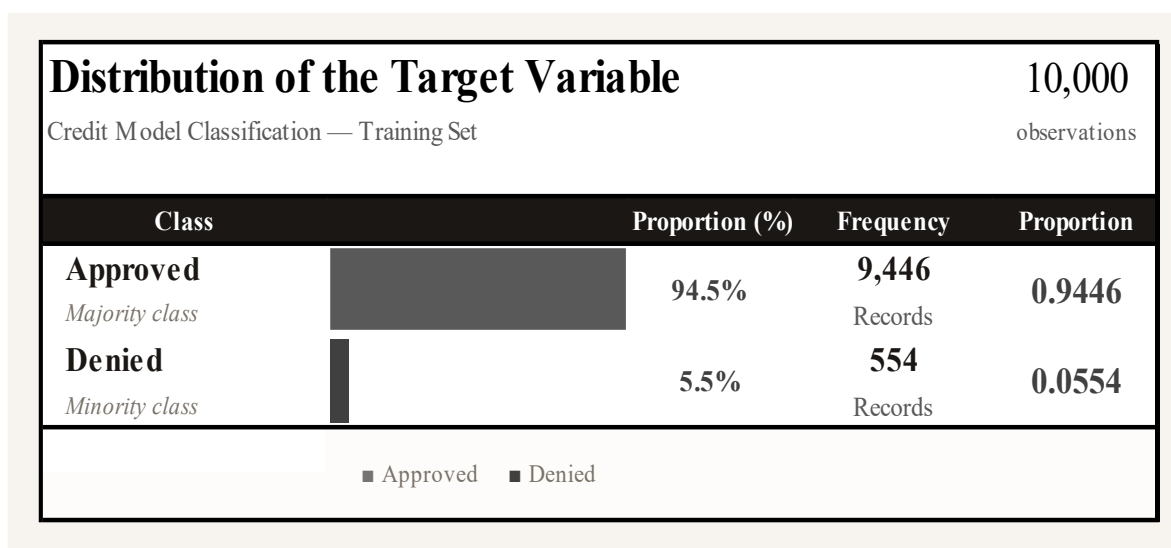


Figure VII : Distribution of the Target Variable

Such imbalance is common in real-world credit risk datasets, since default events are relatively infrequent compared to successful repayments. However, this imbalance can substantially affect model performance if not properly addressed, particularly when using evaluation metrics based solely on overall accuracy. In highly imbalanced datasets, a model may achieve high accuracy simply by predicting the majority class, while failing to correctly identify risky borrowers.

In the context of credit risk modeling, this issue is especially critical because false negatives (predicting a borrower as non-default when they actually default) typically generate higher financial losses than false positives. Therefore, relying exclusively on accuracy would provide a misleading assessment of the model's effectiveness.

For this reason, more robust evaluation metrics such as Precision, Recall, F1-score, ROC-AUC, and PR-AUC were considered throughout the modeling process, as these metrics better capture the model's ability to correctly identify minority default events while maintaining overall predictive performance.

2.1.5 Variable Type

The dataset variables can be grouped into four main categories according to their statistical and analytical characteristics.

Numerical variables (14): *pa_mediana_remesa, pa_media_remesa, pa_desv_remesa, pa_cv_remesa, pa_pendiente_24m, pa_pendiente_norm, pa_fuerza_estacional, pa_cuota_propuesta_mxn, pa_ratio_remesa_cuota, pc_iim_dp2, pc_pct_viv_remesas, pc_pct_viv_emigrantes, pc_imn_2020, pc_escolaridad_promedio_mun, pc_pct_pob_ocupada_2sm, pc_score_regional, pc_sub_capacidad_pago, and pc_sub_intensidad_migratoria.*

Discrete numerical variables (5): *pa_n_meses_sin_envio, pa_max_racha_sin_envio, pa_antiguedad_meses, pb_edad, and pb_n_dependientes.*

Binary variables (3): *pb_genero_F, pb_vivienda_propia, and pb_vivienda_rentada.*

Categorical variables (3): *id_hogar, cve_municipio, and municipio.*

It is important to note that several variables required preprocessing and normalization before the modeling stage. In particular, remittance-related variables exhibited significant scale differences and skewed distributions due to the presence of extreme monetary values and heterogeneous household behavior. Consequently, feature scaling, transformation, and data validation procedures were applied to ensure consistency, numerical stability, and compatibility with machine learning algorithms.

Additionally, identification variables such as *id_hogar*, *cve_municipio*, and *municipio* were excluded from predictive training to avoid information leakage and non-generalizable patterns, while still being retained for traceability and regional analysis purposes.

2.1.6 Data Cleaning

The data cleaning process was limited, as the dataset already exhibited a high level of structure and consistency prior to preprocessing. An initial exploratory analysis confirmed the absence of major missing values, invalid records, or severe inconsistencies that could compromise the modeling process. Likewise, no critical outliers requiring removal were detected, indicating that the dataset was already close to a production-ready analytical environment.

The preprocessing and validation procedures were implemented within a Jupyter Notebook environment, where exploration data analysis (EDA), feature validation, and consistency checks were performed before model development. Initially, a broader ETL (Extract, Transform, Load) pipeline was considered in order to apply extensive transformations, normalization procedures, and automated cleaning routines across the entire dataset. However, after conducting the exploratory analysis, it was determined that the dataset already possessed sufficient quality and structural consistency, making large-scale corrective ETL procedures unnecessary. As a result, the workflow focused primarily on feature engineering, preprocessing optimization, and model development rather than intensive data correction.

The main preprocessing tasks performed were intended to improve modeling consistency and numerical stability. Identification variables such as *id_hogar*, *cve_municipio*, and *municipio* were excluded from the predictive training stage, as they do not provide direct predictive value and could introduce non-generalizable patterns or information leakage into the model. Nevertheless, these variables were retained separately for traceability and regional analysis purposes.

Additionally, several remittance-related variables exhibited skewed distributions and substantial scale differences due to heterogeneous household financial behavior. Consequently, normalization and scaling procedures were applied to improve compatibility with machine learning algorithms and ensure stable model convergence.

Given these conditions, no extensive corrective cleaning procedures were required. The dataset was considered sufficiently dependable for modeling purposes, allowing the project

to concentrate on feature engineering, class imbalance handling, hyperparameter optimization, and predictive model validation.

2.1.7 Preprocessing

The preprocessing stage was designed to transform the raw remittance-based dataset into a structured and machine learning-ready format. The preprocessing pipeline was implemented in Python using a modular architecture within Jupyter Notebook, allowing reproducibility, scalability, and clear separation between data preparation and modeling stages.

The first step of the preprocessing workflow consisted of separating the target variable (*default_12m*) from the predictor variables. Subsequently, non-predictive identification fields were removed from the feature matrix in order to reduce potential information leakage and avoid the incorporation of non-generalizable identifiers into the model training process.

Next, categorical variables were transformed using one-hot encoding through the *pd.get_dummies()* procedure with the *drop_first=True* parameter to avoid multicollinearity issues. Although the dataset was primarily composed of numerical variables, this transformation ensured compatibility across different machine learning algorithms and standardized the feature representation format.

Additionally, preprocessing procedures included validation mechanisms to verify the structural consistency of the dataset after transformations and encoding. These checks ensured that the final dataset maintained the expected format prior to model training and experimentation.

The preprocessing stage was intentionally designed to leave the dataset fully prepared for subsequent feature engineering, feature analysis, and feature selection procedures. This modular structure allowed new variables, derived indicators, and transformed features to be incorporated efficiently during later stages of the modeling workflow without requiring major modifications to the preprocessing pipeline.

Finally, the processed dataset was divided into training and testing subsets using a stratified train-test split procedure. A test size of 25% was used while preserving the original distribution of the target variable through stratification. In addition, a fixed random seed

(*random_state* = 42) was established to ensure reproducibility and consistency across experiments.

2.2 Exploratory Feature Analysis

To ensure that the development of the remittance-based credit model did not remain purely a statistical exercise, financial, behavioral, sociodemographic, and regional variables were incorporated to objectively evaluate household creditworthiness and estimate the probability of default. This methodology follows the logic of modern alternative credit scoring frameworks, where non-traditional financial indicators are used to model repayment capacity and support lending decisions for underserved populations without extensive banking histories.

Remittance Stability and Financial Capacity

This category includes variables associated with the stability, consistency, and reliability of remittance inflows, which constitute the primary source of income for many households in the dataset. Variables such as *pa_media_remesa*, *pa_mediana_remesa*, *pa_cv_remesa*, *pa_desv_remesa*, and *pa_ratio_remesa_cuota* provide insight into the household's financial capacity and ability to sustain loan payments over time.

In particular, the remittance-to-installment ratio serves as a direct approximation of repayment capacity, while volatility indicators such as the coefficient of variation and standard deviation help identify unstable income behavior. Higher remittance instability may indicate increased financial vulnerability and therefore a greater probability of default.

Remittance Behavioral Patterns

This dimension captures temporal and behavioral characteristics associated with remittance activity. Variables such as *pa_n_mesese_sin_envio*, *pa_max_racha_sin_envio*, *pa_pendiente_24m*, *pa_pendiente_norm*, *pa_fuerza_estacional*, and *pa_antiguedad_mesese* help characterize the long-term consistency of remittance flows and the household's historical financial behavior.

For example, prolonged periods without remittance inflows or negative remittance trends may indicate income deterioration or financial instability. Conversely, households with longer remittance histories and stronger seasonal consistency may demonstrate more stable economic behavior and lower default risk. These variables function similarly to behavioral credit indicators used in traditional banking systems, where transaction consistency and historical payment patterns are considered strong predictors of future repayment behavior.

Household Sociodemographic Characteristics

This category includes demographic and household-level variables that may indirectly reflect economic resilience and repayment reliability. Variables such as *pb_edad*, *pb_n_dependientes*, *pb_escolaridad_ord*, *pb_vivienda_propia*, and *pb_vivienda_rentada* help characterize the socioeconomic profile of the beneficiary household.

Educational attainment may serve as a proxy for financial literacy and long-term income stability, while housing conditions can reflect household wealth accumulation and financial security. Additionally, the number of economic dependents may influence disposable income availability and repayment capacity under financial stress conditions.

Regional Socioeconomic and Migration Indicators

This dimension incorporates municipal-level variables associated with migration intensity, socioeconomic vulnerability, and regional economic conditions. Variables such as *pc_iim_dp2*, *pc_pct_viv_remesas*, *pc_pct_viv_emigrantes*, *pc_imn_2020*, *pc_score_regional*, and *pc_sub_capacidad_pago* provide contextual information regarding the economic environment surrounding each household.

These indicators are particularly relevant in remittance-based credit systems because regional migration dynamics and local economic conditions strongly influence the stability of remittance flows and household financial behavior. Municipalities with stronger migration networks and higher remittance penetration may exhibit different repayment dynamics compared to regions with weaker migration dependence or higher marginalization levels.

The feature analysis stage allowed the project to integrate financial behavior, household characteristics, and regional socioeconomic context into a unified analytical framework for

credit risk modeling. This multidimensional approach improves the interpretability of the model and provides a more realistic approximation of borrower repayment behavior in alternative lending environments.

Additionally, a correlation analysis was conducted to identify potential multicollinearity issues among the explanatory variables.

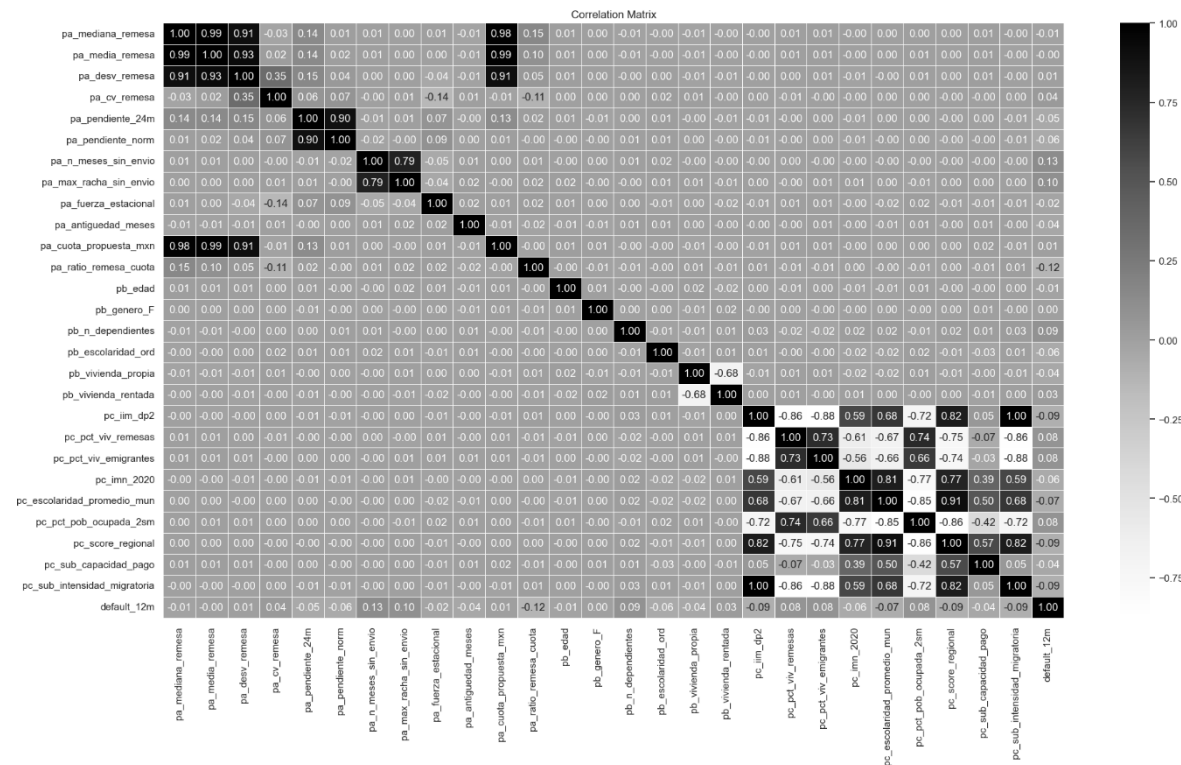


Table III : Correlation Matrix of all variables

The initial correlation matrix revealed the presence of several highly correlated variables, particularly within the remittance-related and regional socioeconomic indicators. Strong correlations were observed between variables associated with remittance magnitude and dispersion, as well as among multiple socioeconomic indicators derived from similar regional information sources.

High multicollinearity can negatively affect model interpretability, increase redundancy, and reduce the stability of certain machine learning algorithms. For this reason, a feature

reduction process was performed in order to retain the most representative variables while removing highly collinear features that provided overlapping information.

For example, variables such as *pa_mediana_remesa*, *pa_desv_remesa*, and *pa_cuota_propuesta_mxn* exhibited strong correlations with other remittance-related metrics and were therefore excluded from the final modeling set. Similarly, several regional indicators presented substantial correlation coefficients due to their shared socioeconomic structure, leading to the retention of only the most representative regional variables for the final model.

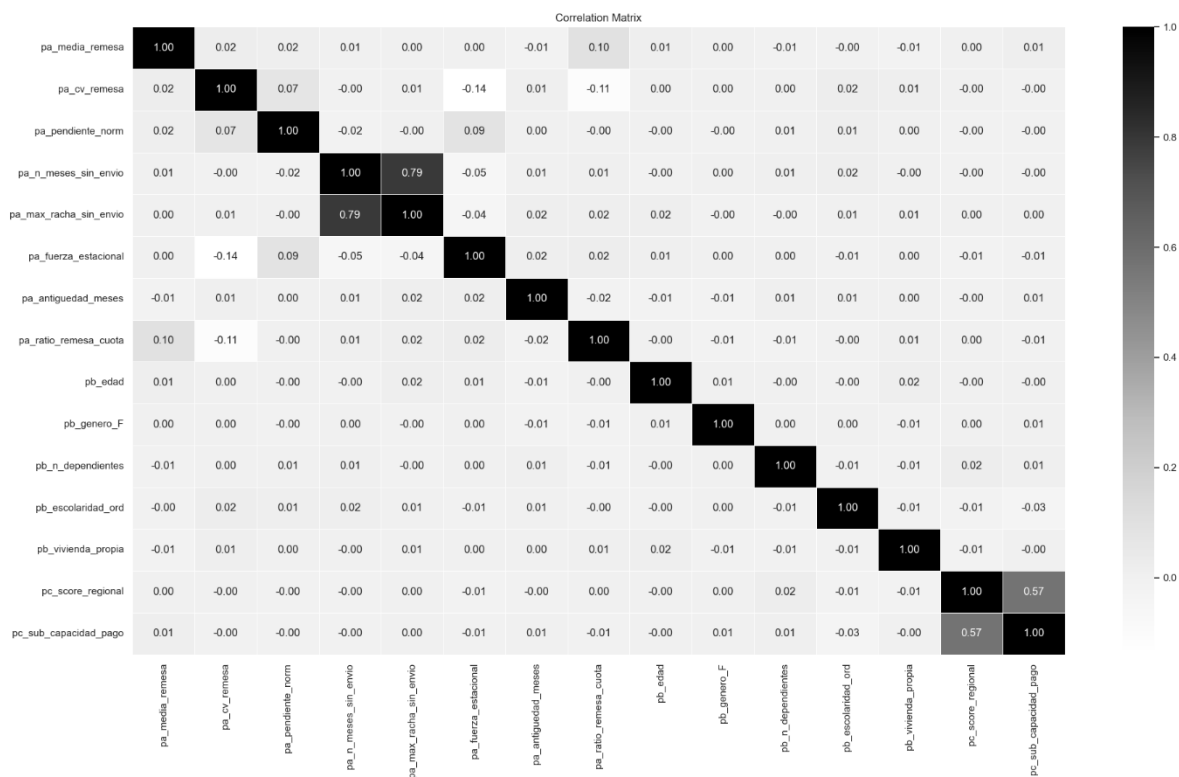


Table IV : Correlation Matrix of clean variables

After the feature reduction stage, a second correlation analysis confirmed that the remaining variables exhibited lower levels of multicollinearity, resulting in a more stable and interpretable feature space for the machine learning models.

2.3 Feature Importance & Explainability

To mitigate potential human bias in manual variable selection and reduce the inclusion of predictors that may introduce noise into the model, a feature importance filtering process was implemented using the gradient boosting decision tree algorithm XGBoost as an additional validation mechanism for variable relevance.

In remittance-based credit risk modeling, ensemble learning techniques such as XGBoost are particularly valuable due to their ability to capture nonlinear relationships and complex interactions among financial, behavioral, and regional socioeconomic variables that may not be identified through traditional statistical methods. Furthermore, XGBoost provides a quantitative measure of feature importance based on gain, which evaluates the improvement in predictive performance generated by each variable when used in tree-splitting decisions during the boosting process.

By iteratively evaluating the contribution of each predictor throughout model training, this methodology enables the identification of variables with the highest explanatory power in relation to household default behavior. As illustrated in Figure VII, variables associated with remittance interruption patterns and repayment capacity exhibited the highest importance scores. In particular, `pa_n_meses_sin_envio`, `pa_ratio_remesa_cuota`, and `pa_max_racha_sin_envio` emerged as the most influential predictors, indicating that remittance consistency and household payment capacity play a significant role in estimating credit risk.

Additionally, regional socioeconomic indicators such as *`pc_score_regional`* and *`pc_sub_capacidad_pago`*, together with household-level variables including *`pb_n_dependientes`*, *`pb_escolaridad_ord`*, and *`pb_vivienda_propia`*, also demonstrated meaningful predictive contribution. These results suggest that both financial behavior and socioeconomic context are important components in the construction of alternative credit scoring systems for remittance-dependent populations.

Conversely, variables with lower feature importance scores, while still potentially informative, demonstrated comparatively smaller marginal contributions to the predictive

framework. This process provided an additional layer of statistical validation for feature selection, improving model interpretability and helping ensure that the final predictive model was constructed using the most informative and representative household characteristics.

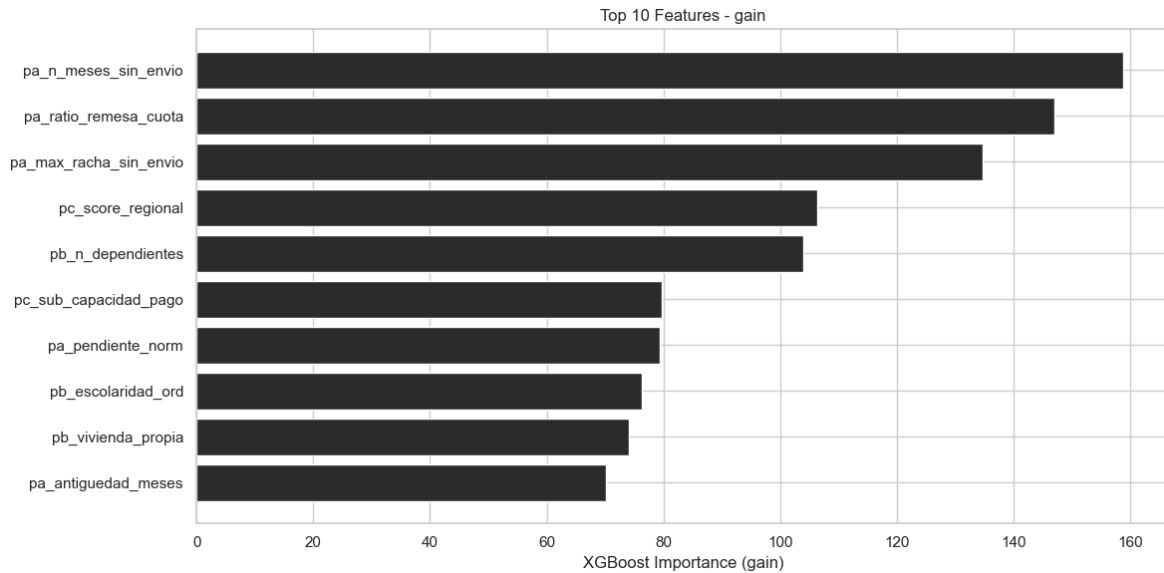


Figure VIII : Top 10 Most Important Features Based on XGBoost Gain

Although the results presented in the previous figure provide a useful indication of which variables may be suitable for inclusion in the final model, feature importance metrics only measure the statistical contribution of each variable and do not capture the direction or economic interpretation of the relationship with the target variable. In other words, a high importance score does not necessarily imply a financially meaningful relationship with default behavior.

For this reason, additional analysis was conducted to evaluate the economic consistency and interpretability of the selected predictors. Variables related to remittance stability, payment capacity, and regional socioeconomic conditions were further examined to ensure that their relationship with default risk remained financially coherent within the context of remittance-based credit modeling.

2.3.1 XGBoost Hyperparameter Optimization

XGBoost Hyperparameters		
Parameter	Value	Description
objective	binary:logistic	Defines a binary classification objective and outputs default probabilities.
n_estimators	300	Number of boosting trees used during training.
max_depth	3	Maximum depth of each decision tree, controlling model complexity.
learning_rate	0.01	Shrinks the contribution of each tree to improve generalization.
subsample	0.8	Fraction of training observations randomly sampled for each boosting iteration.
colsample_bytree	0.8	Fraction of features randomly sampled for each tree.
eval_metric	logloss	Evaluation metric used to measure probabilistic prediction error.
random_state	42	Fixed seed used to ensure reproducibility of results.
scale_pos_weight	scale_pos_weight	Adjusts class imbalance by increasing the weight of minority default observations.

Table V : Hyperparameters Used for XGBoost

An XGBoost classifier was trained using the hyperparameters presented below. These parameters were selected to balance predictive performance, model complexity, and feature importance stability while reducing the risk of overfitting through controlled learning and subsampling techniques.

The results indicate that a subset of variables contributes more significantly to the prediction of household default behavior. In particular, variables associated with remittance interruption patterns, repayment capacity, and regional socioeconomic conditions demonstrated the highest predictive contribution. These predictors were therefore considered strong candidates for the final model,

as they provided the highest information gain and explanatory power in estimating default_{12m}.

2.4 SHAP Analysis

To interpret the relationships identified by the model, the Shapley Additive Explanations (SHAP) framework was implemented as an interpretability tool for the selected variables. This technique decomposes the predictions generated by the XGBoost model into the additive contribution of each feature, allowing the magnitude and direction of each variable's impact on the probability of household default to be observed for every individual prediction (Gramegna & Giudici, 2021).

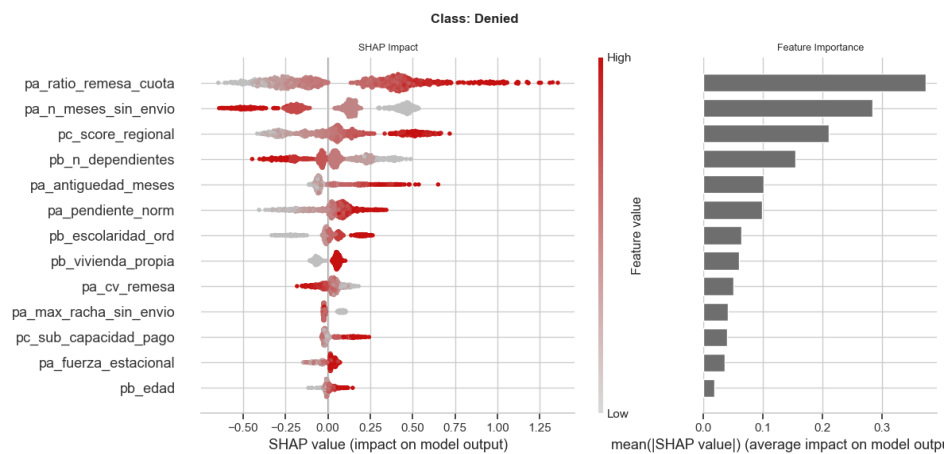


Figure IX. a : SHAP Feature Importance and Impact Analysis — Denied Class

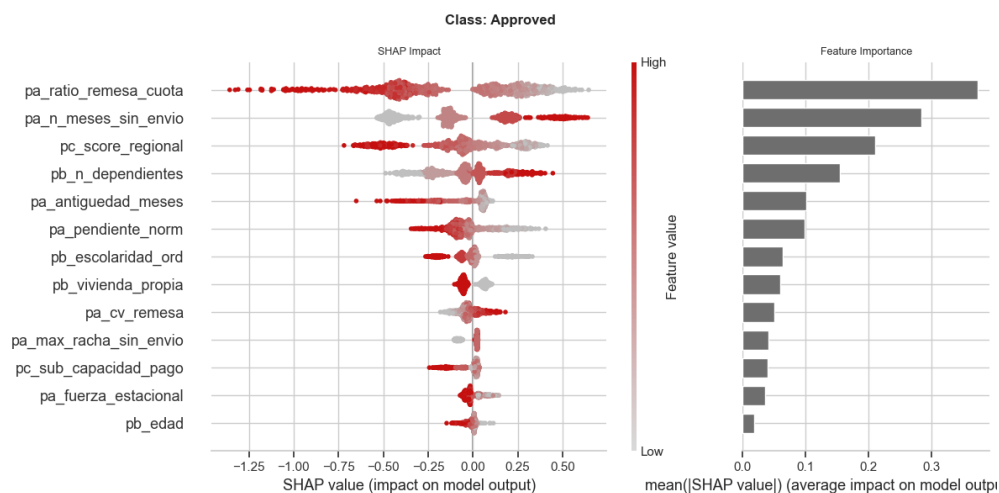


Figure IX. b : SHAP Feature Importance and Impact Analysis — Approved Class

As shown in the SHAP value distributions for both the Non-Default and Default classes (Figures IX. and IX.b), variables associated with remittance stability and repayment capacity exhibit the strongest directional influence on model predictions. In particular, *pa_ratio_remesa_cuota* demonstrates a significant contribution to the classification outcome. Higher values of this variable contribute positively toward the Non-Default classification, indicating that households with stronger remittance-to-installment capacity are less likely to default. Conversely, lower values shift predictions toward the Default category, reflecting reduced repayment capability.

Similarly, variables such as *pa_n_meses_sin_envio* and *pa_max_racha_sin_envio* show strong behavioral effects on the probability of default. Higher values for these variables increase the likelihood of Default predictions, suggesting that prolonged interruptions in remittance inflows are associated with greater financial instability and elevated credit risk.

Regional socioeconomic indicators, particularly *pc_score_regional* and *pc_sub_capacidad_pago*, also demonstrate meaningful influence on the model's predictions. Households located in regions with stronger socioeconomic conditions and higher payment capacity scores tend to contribute positively toward the Non-Default classification, while more vulnerable regional conditions shift predictions toward Default outcomes.

Additionally, sociodemographic variables such as *pb_n_dependientes*, *pb_escolaridad_ord*, and *pb_vivienda_propia* contribute to the classification process, although with comparatively lower impact. These variables provide complementary information regarding household financial resilience, economic stability, and long-term repayment reliability within the alternative credit scoring framework.

2.5 Feature Selection

After completing the theoretical and empirical validation stages through correlation analysis, feature importance evaluation, and SHAP interpretability analysis, a final selection of variables was determined for the construction of the remittance-based credit scoring model. This selection was designed to preserve a balanced representation of the main household risk dimensions while ensuring both statistical relevance and financial interpretability within the alternative credit assessment framework.

Remittance Stability and Repayment Capacity

The following variables were selected:

- *pa_media_remesa*
- *pa_cv_remesa*

- *pa_ratio_remesa_cuota*

These variables were incorporated to capture the magnitude, stability, and repayment capacity associated with household remittance inflows. Remittances represent the primary source for many households in the dataset; therefore, evaluating both the consistency and sufficiency of these inflows is fundamental for estimating repayment capability.

In particular, *pa_ratio_remesa_cuota* was selected because it directly measures the relationship between household remittance income and the proposed loan installment, functioning as a proxy for debt affordability. Similarly, *pa_cv_remesa* was included to evaluate remittance volatility, since unstable income flows may increase financial vulnerability and repayment uncertainty.

Remittance Behavioral Patterns

The following variables were selected:

- *pa_pendiente_norm*
- *pa_n_meses_sin_envio*
- *pa_max_racha_sin_envio*
- *pa_fuerza_estacional*
- *pa_antiguedad_meses*

These variables capture temporal and behavioral characteristics associated with remittance activity and financial consistency over time. Variables related to interruptions in remittance inflows were particularly relevant because prolonged periods without transfers may indicate economic instability or deterioration in household income conditions.

Additionally, *pa_antiguedad_meses* was incorporated to reflect the length of remittance history available for each household. Longer remittance histories may provide stronger evidence regarding long-term financial consistency and income reliability. Seasonal and trend-related variables were also included because recurring remittance behavior patterns may contain valuable information about future financial stability.

Household Sociodemographic Characteristics

The following variables were selected:

- *pb_edad*
- *pb_genero_F*
- *pb_n_dependientes*
- *pb_escolaridad_ord*
- *pb_vivienda_propia*

These variables were incorporated to represent household demographic and socioeconomic stability. Educational attainment and home ownership may function as indirect indicators of long-term financial resilience, while the number of dependents may influence household financial pressure and disposable income availability.

Additionally, age was included as a proxy for financial maturity and economic stability, while gender-related variables were retained to evaluate whether demographic segmentation contributed additional predictive information within the alternative credit framework.

Regional Socioeconomic Indicators

The following variables were selected:

- *pc_score_regional*
- *pc_sub_capacidad_pago*

These variables capture the broader socioeconomic environment surrounding each household. Regional economic conditions play an important role in remittance-based credit systems because local vulnerability and payment capacity may directly influence financial stability and default behavior.

The incorporation of regional indicators allows the model to account not only for household-level financial behavior but also for contextual economic conditions that may affect

repayment performance. This multidimensional structure improves the robustness and interpretability of the final credit risk model.

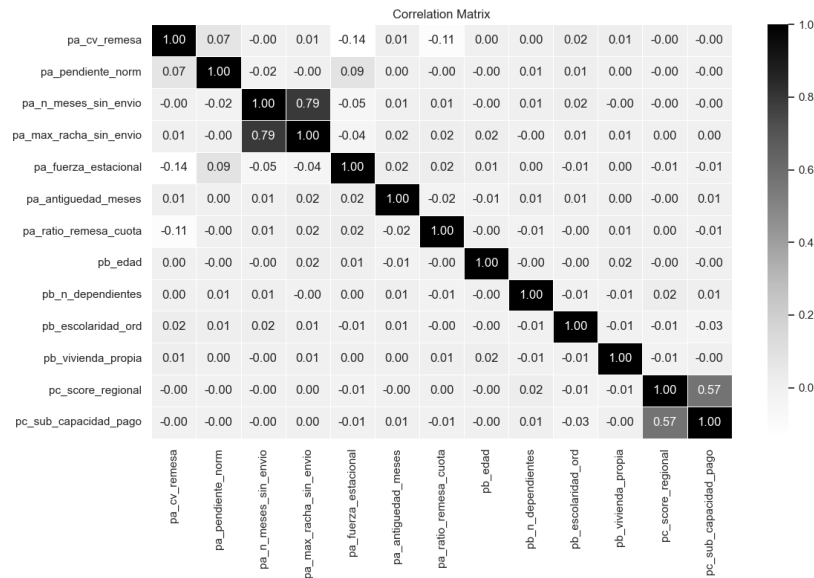


Table VI : Correlation Matrix of Selected variables

Finally, the correlation matrix presented in “table vi” was used as an additional validation tool during the feature selection process to assess potential multicollinearity among the final predictor variables. The results indicate low to moderate pairwise correlations across the selected features, suggesting that the final variable set does not exhibit excessive linear dependence and that each predictor contributes distinct explanatory information to the model.

Although moderate relationships are observed between certain variables, most notably between *pa_n_mesess_sin_envio* and *pa_max_racha_sin_envio*, as well as between *pc_score_regional* and *pc_sub_capacidad_pago*, these correlations remain within acceptable ranges and are economically justifiable given the natural relationship between remittance interruption behavior and regional socioeconomic conditions.

The correlation analysis supports the robustness of the selected feature set by confirming that the final model preserves explanatory diversity while minimizing redundancy among predictors.

3. Model Architecture & System Design

3.1 Architecture

3.1.1 Project Structure

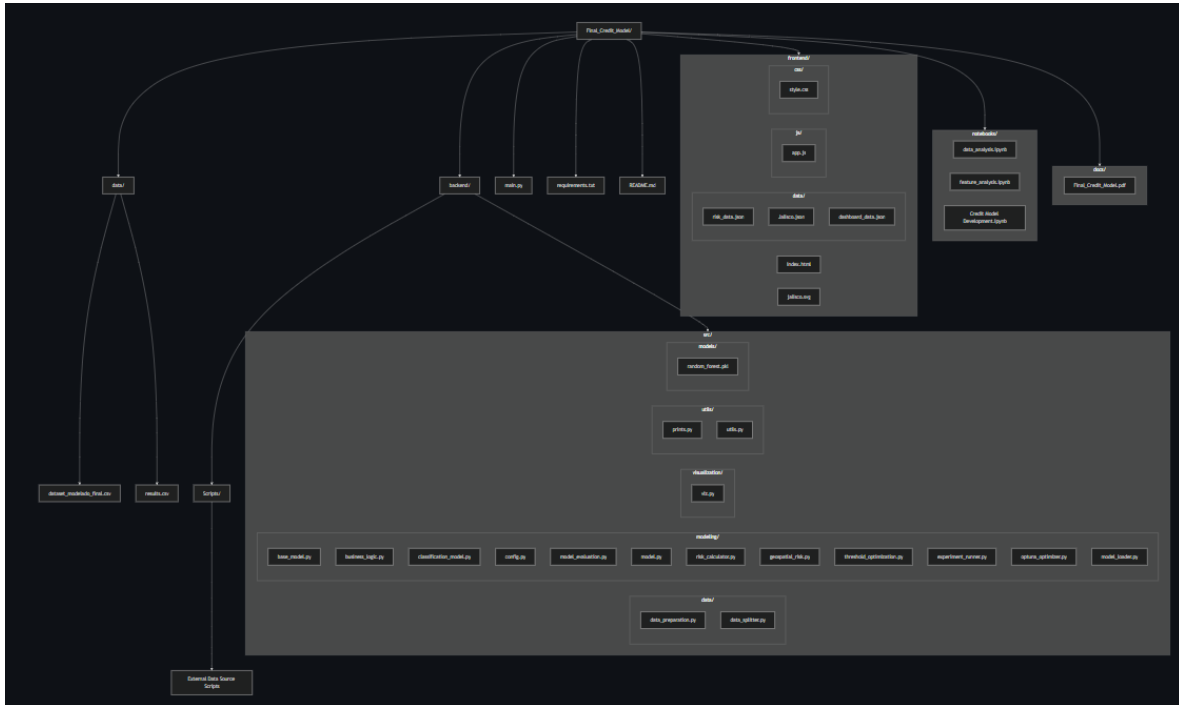


Figure X : Project Architecture and Modular File Structure of the Credit Risk Model

3.1.2 Object-Oriented Design

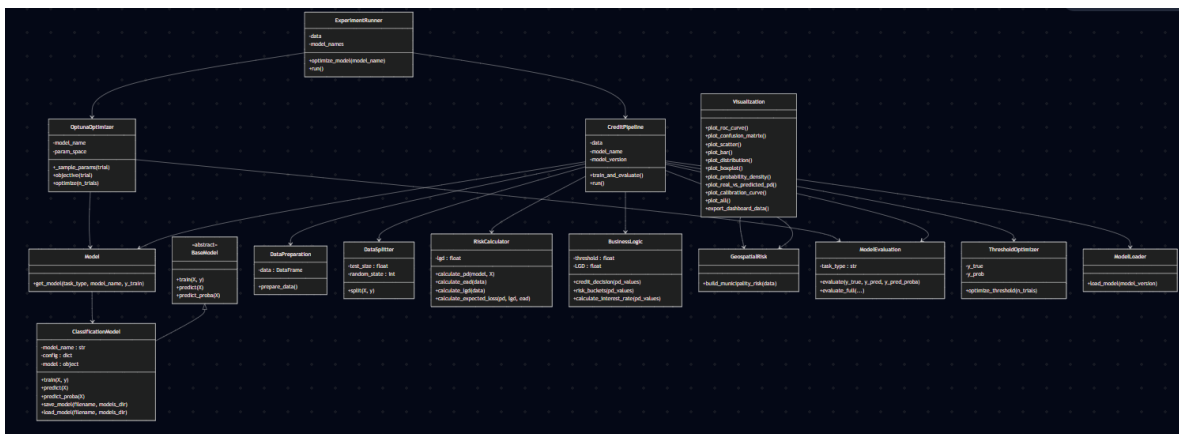


Figure XI : Object-Oriented Design of the Credit Risk Framework

3.1.3 Credit Risk System Architecture

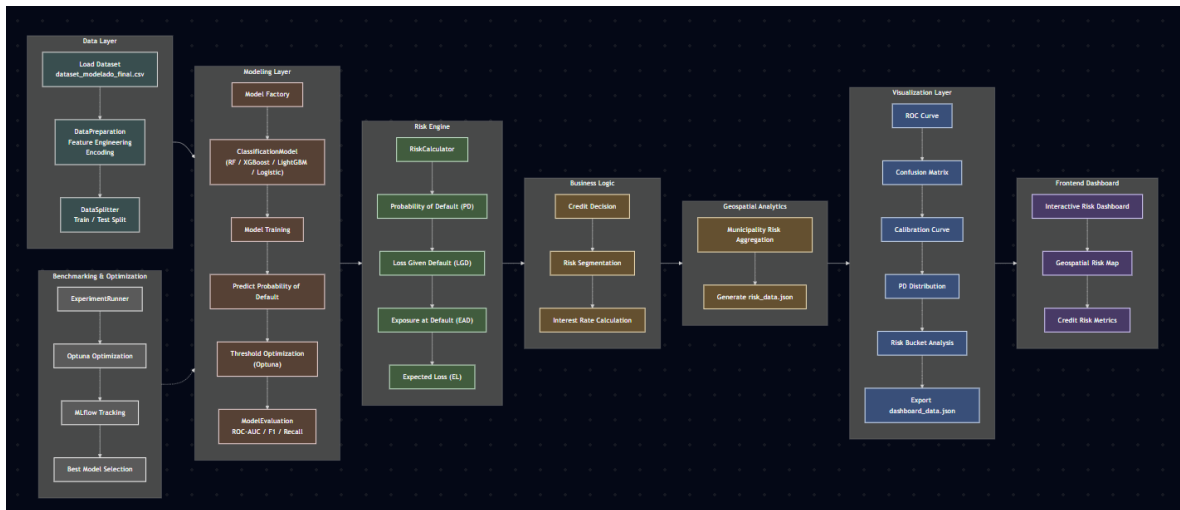


Figure XII : System Architecture of the End-to-End Credit Risk Modeling and Deployment Framework

3.2 System Design

3.2.1 ML System Pipeline Overview

Following the feature engineering and feature selection stages, a complete end-to-end machine learning framework was developed to operationalize the proposed remittance-based credit risk model into a scalable and deployment-ready analytical platform. Rather than limiting the project to standalone predictive experimentation, the objective was to construct an integrated system capable of supporting data processing, experimentation, model evaluation, risk analytics, and web-based visualization within a unified architecture.

The framework was implemented using a modular pipeline architecture developed primarily in Python and integrated with machine learning and analytical libraries including Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, MLflow, and Optuna. The system automates the complete modeling lifecycle beginning with raw data ingestion and ending with deployment-oriented analytical outputs and geospatial visualization components.

As illustrated in the system architecture diagram, the workflow begins with multiple external remittance, demographic, financial, and socioeconomic data sources. These datasets are processed through a dedicated feature engineering pipeline responsible for temporal disaggregation, municipality-level aggregation, synthetic household generation, regional score construction, and borrower-level target generation. The resulting outputs are

consolidated into the final structured dataset used throughout the experimentation framework.

Once generated, the dataset enters the machine learning pipeline, where preprocessing, train-test splitting, feature validation, encoding, and scaling procedures are executed automatically. The processed data is subsequently integrated into the experimentation framework, allowing multiple machine learning algorithms to be trained, optimized, and evaluated dynamically under a unified structure.

To support experimentation reproducibility and centralized model management, the framework was integrated with MLflow, enabling systematic experiment tracking, parameter versioning, metric logging, and model lifecycle monitoring. Additionally, Optuna-based hyperparameter optimization was incorporated to automate parameter search and improve model selection efficiency across candidate algorithms.

The system was intentionally designed using object-oriented and modular architecture to facilitate scalability, maintainability, and future extensibility. This structure allows additional machine learning models, feature engineering modules, and regional datasets to be incorporated into the framework without requiring substantial modifications to the existing pipeline.

Beyond predictive modeling, the platform also integrates business-oriented credit risk components including Probability of Default (PD), Loss Given Default (LGD), Exposure at Default (EAD), expected loss estimation, borrower segmentation, and risk-adjusted pricing mechanisms. This transforms the framework from a traditional classification pipeline into a broader financial risk analytics system capable of supporting underwriting and portfolio-level decision-making processes.

Finally, the analytical outputs generated throughout the pipeline are integrated into a web-based visualization platform deployed through Vercel. The deployed system centralizes geospatial analytics, model monitoring, experimentation results, and borrower risk segmentation within an interactive interface accessible through GitHub integration or public deployment links. This deployment architecture converts the project from a standalone

academic modeling exercise into a deployable end-to-end machine learning application with practical scalability and operational usability considerations.

Modular Modeling Architecture

To ensure scalability, maintainability, and experimentation flexibility, the machine learning framework was designed using a modular object-oriented architecture that allows multiple predictive algorithms and analytical components to operate under a unified structure. The system was intentionally developed to support extensibility and reproducibility while maintaining consistent integration across the full modeling pipeline.

At the core of the framework, an abstract modeling structure was implemented to standardize the training, prediction, and probability estimation processes across all candidate machine learning models. This architecture allows different algorithms to be dynamically integrated into the experimentation pipeline without requiring modifications to the broader workflow.

The framework currently supports multiple classification models including Logistic Regression, Random Forest, XGBoost, and LightGBM, enabling systematic benchmarking and comparative evaluation under identical preprocessing and validation conditions. By centralizing model implementation under a shared architecture, the system facilitates consistent experimentation and improves maintainability throughout the development process.

To simplify experimentation management, model hyperparameters and configuration settings were centralized through dedicated configuration modules, allowing parameter control, optimization boundaries, and model initialization settings to remain fully standardized across experiments. This structure improves reproducibility while simplifying integration with the Optuna optimization framework and the MLflow experiment tracking system.

Additionally, the architecture incorporates reusable evaluation and visualization modules capable of generating classification metrics, ROC curves, confusion matrices, probability distributions, SHAP interpretability outputs, feature importance analysis, and borrower risk segmentation analytics. These modules allow statistical validation and business-oriented interpretation to operate cohesively within the same analytical environment.

The modular design also improves deployment readiness by separating preprocessing, experimentation, evaluation, business logic, and visualization responsibilities into independent components. This separation facilitates future scalability and allows additional predictive models, regional datasets, or frontend analytical modules to be incorporated into the framework with minimal structural modifications.

Business Logic and Risk Integration

Beyond predictive classification, the framework incorporates a dedicated business logic and risk integration layer designed to transform machine learning outputs into actionable financial decision-making components. This allows the system to extend beyond binary classification and operate as a broader credit risk analytics framework capable of supporting underwriting, pricing, and portfolio-level segmentation processes.

The business logic module was implemented through modular Python architecture integrating borrower approval rules, Probability of Default (PD) estimation, risk bucket assignment, expected loss calculation, and risk-adjusted interest rate construction.

Following model training, the predicted probabilities generated by the selected machine learning model are interpreted as borrower-level PD estimates. These probabilities serve as the central risk indicator throughout the analytical framework and are subsequently integrated into multiple financial and business-oriented calculations.

To support credit decision-making, the framework incorporates a threshold-based approval mechanism in which borrowers with PD values below a predefined cutoff are classified as approved, while higher-risk borrowers are rejected according to the institution's internal risk tolerance strategy.

Additionally, borrowers are segmented into Low, Medium, and High-Risk categories through a probability-based bucket structure derived directly from the predicted PD distributions. This segmentation framework supports portfolio monitoring, borrower classification, and risk concentration analysis across different borrower profiles.

The framework also integrates traditional credit risk metrics commonly used within financial institutions. Exposure at Default (EAD) is estimated using the proposed borrower installment

amount as a proxy for annualized exposure, while Loss Given Default (LGD) is incorporated as a fixed recovery assumption. These components are combined with the estimated PD values to compute Expected Loss (EL) through the standard credit risk formulation:

$$EL = PD \cdot LGD \cdot EAD$$

This structure allows borrower-level projected monetary losses to be estimated and incorporated into portfolio-level financial risk analysis.

To improve the practical applicability of the framework, a risk-adjusted pricing module was also implemented. The pricing structure incorporates borrower-specific risk premiums derived from predicted PD values together with operational costs, funding spreads, capital costs, and profit margin assumptions. Under this framework, borrowers with higher predicted risk levels systematically receive higher modeled interest rates, replicating standard risk-based lending practices.

The integration of predictive modeling, financial risk metrics, borrower segmentation, approval rules, and pricing logic allows the framework to operate as a unified credit risk system capable of supporting both statistical modeling and practical lending decision processes.

3.2.2 Model Evaluation

To assess the predictive performance of the proposed credit risk classification model, several evaluation metrics were employed, including Accuracy, Precision, Recall, F1-Score, and ROC-AUC, as recommended in classification modeling literature (James et al., 2021). These metrics provide complementary perspectives regarding the model's predictive capability and discriminatory performance.

Accuracy

Measures the proportion of correctly classified observations relative to the total number of observations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP denotes true positives, TN true negatives, FP false positives, and FN false negatives.

Precision

Evaluates the proportion of positive predictions that were correctly identified:

$$Precision = \frac{TP}{TP + FP}$$

This metric reflects the reliability of the model when predicting default cases.

Recall

Recall, also referred to as sensitivity, measures the proportion of actual positive cases correctly identified by the model:

$$Recall = \frac{TP}{TP + FN}$$

This metric is especially important in credit risk applications, as it indicates the model's ability to identify borrowers who are likely to default.

F1-Score

Represents the harmonic mean between precision and recall, balancing both metrics into a single indicator:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

This metric is useful when evaluating classification problems with potential imbalance between classes.

ROC-AUC

Finally, ROC-AUC (Receiver Operating Characteristic – Area Under the Curve) measures the model's overall ability to discriminate between positive and negative classes across all

classification thresholds. It is based on the relationship between the true positive rate and false positive rate:

$$TPR = \frac{TP}{TP + FN}$$
$$FPR = \frac{FP}{FP + TN}$$

Higher ROC-AUC values indicate stronger discriminatory power, where 0.50 reflects random classification and 1.00 indicates perfect predictive separation.

Model Benchmarking and Selection

To identify the most suitable predictive framework for remittance-based credit risk assessment, multiple machine learning algorithms were evaluated under a unified experimental environment. The objective of this benchmarking process was to compare predictive performance, robustness, interpretability, and deployment feasibility across different classification approaches while maintaining consistency in preprocessing, feature engineering, and evaluation methodology.

The candidate algorithms selected for experimentation included Logistic Regression, Random Forest, XGBoost, and LightGBM. These models were chosen because they represent different levels of complexity and modeling capabilities commonly used in modern credit risk systems. Logistic Regression was included as a traditional baseline model widely adopted in financial institutions due to its simplicity and interpretability, while ensemble tree-based methods such as Random Forest, XGBoost, and LightGBM were incorporated because of their ability to capture nonlinear relationships and complex interactions among behavioral and socioeconomic variables.

To ensure fairness and reproducibility across experiments, all models were trained using the same preprocessing pipeline, train-test split configuration, selected feature set, and evaluation metrics. Additionally, hyperparameter optimization was performed using Optuna, allowing automated exploration of parameter configurations through an efficient search process designed to maximize predictive performance while reducing overfitting risk.

The experimentation framework was integrated with MLflow to support centralized experiment tracking, parameter logging, metric management, and model versioning. This architecture enabled systematic comparison between candidate algorithms and ensured full reproducibility of the modeling process.

The evaluation process focused not only on predictive accuracy, but also on metrics particularly relevant for imbalanced credit risk classification problems, including ROC-AUC, Precision-Recall performance, F1-score, recall sensitivity, calibration behavior, and business interpretability. This multidimensional evaluation strategy ensured that the final selected model achieved strong statistical performance while remaining economically coherent and operationally suitable for deployment within the proposed ML system.

3.3 Business-Oriented Model Comparison

Beyond predictive performance, the candidate models were also evaluated according to operational and business-oriented considerations relevant for real-world deployment environments. Factors such as interpretability, computational complexity, training efficiency, nonlinear learning capability, and suitability for imbalanced credit risk classification were considered during the final evaluation process.

Table VII summarizes the qualitative comparison across the evaluated machine learning frameworks.

Model Comparison					
Model	Interpretability	Training Speed	Non-Linearity	Imbalanced Data Handling	Deployment Complexity
Logistic Regression	High	Very Fast	Low	Medium	Low
Random Forest	Medium	Medium	High	High	Medium
XGBoost	Medium	Slow	Very High	Very High	High
LightGBM	Medium	Fast	Very High	Very High	Medium

Table VII : Business-Oriented Comparison of Candidate Machine Learning Models

The comparison indicates that ensemble boosting methods provide stronger nonlinear learning capability and improved minority class sensitivity, although they introduce higher computational complexity and lower interpretability.

Conversely, Logistic Regression provides significantly faster training, simpler deployment, and greater model transparency while still maintaining competitive predictive performance. These characteristics are especially valuable in credit risk environments where interpretability, stability, and operational simplicity remain important considerations for production deployment and financial decision-making processes.

3.3.1 Hyperparameter Optimization Framework

To ensure a fair and systematic comparison across candidate machine learning algorithms, an automated hyperparameter optimization framework was implemented using Optuna integrated within the experimental pipeline. The objective of this framework was to identify parameter configurations that maximize predictive performance while reducing overfitting and improving model generalization.

The optimization process was designed under a modular architecture where each machine learning model was associated with a predefined hyperparameter search space stored within a centralized configuration module. This structure improved maintainability, reproducibility, and scalability across experiments, allowing all candidate algorithms to be optimized under a consistent framework.

During each optimization trial, Optuna sampled a new parameter configuration which was then used to train and evaluate the selected classification model. The optimization objective was based primarily on ROC-AUC performance due to the imbalanced nature of the credit risk dataset. Additionally, threshold optimization procedures were incorporated to improve F1-score performance and enhance minority default detection capability.

Model Configuration	
Model	Selected Configuration
Logistic Regression	$C \in [0.001, 10]$
Random Forest	$n_estimators \in [100, 500]$, $max_depth \in [3, 15]$, $min_samples_split \in [2, 50]$, $min_samples_leaf \in [1, 20]$
XGBoost	$n_estimators \in [100, 500]$, $max_depth \in [3, 10]$, $learning_rate \in [0.01, 0.2]$, $subsample \in [0.6, 1.0]$, $colsample_bytree \in [0.6, 1.0]$, $min_child_weight \in [1, 15]$, $gamma \in [0, 5]$, $reg_alpha \in [0, 5]$, $reg_lambda \in [0, 10]$
LightGBM	$n_estimators \in [100, 500]$, $learning_rate \in [0.01, 0.2]$, $max_depth \in [3, 10]$, $num_leaves \in [20, 100]$, $subsample \in [0.6, 1.0]$, $colsample_bytree \in [0.6, 1.0]$, $reg_alpha \in [0, 5]$, $reg_lambda \in [0, 10]$

Table VIII : Hyperparameter Search Space Configuration for Candidate Models

To support experiment reproducibility and centralized model management, the optimization workflow was integrated with MLflow, enabling automatic logging of metrics, hyperparameters, experiment runs, and trained model artifacts. This framework facilitated efficient benchmarking across multiple candidate algorithms while maintaining consistency throughout the experimentation process.

Table VIII summarizes the primary hyperparameters optimized for each candidate model during the experimentation stage.

The optimization framework allowed the project to systematically evaluate alternative machine learning configurations while balancing predictive performance, model complexity, and generalization stability. This process established the foundation for the benchmarking and final model selection stages presented in the following sections.

3.4 Experimental Results

Following the hyperparameter optimization stage, the candidate machine learning models were evaluated using multiple classification metrics to assess predictive performance under the imbalanced credit risk setting.

All models were trained using the same preprocessing pipeline, feature set, train-test split configuration, and optimization framework to ensure consistency across experiments. Performance evaluation focused primarily on ROC-AUC, Precision, Recall, F1-score, and Precision-Recall behavior due to the importance of minority default detection in credit risk modeling. The following sections present the benchmarking results, performance comparisons, probability of default analysis, and risk segmentation outputs used for final model selection.

Benchmark Performance Comparison – Version 1

The first experimentation stage evaluated the baseline configurations of the candidate machine learning models using the initial hyperparameter sets generated through the Optuna optimization framework.

Model Performance - V1				
Model Version	ROC-AUC	Precision	Recall	F1-Score
logistic_v1	0.786	0.2335	0.3841	0.2904
random_forest_v1	0.7774	0.1863	0.4348	0.2609
xgboost_v1	0.7724	0.1663	0.558	0.2562
lightgbm_v1	0.7739	0.1739	0.4348	0.2484

Table IX .a. Model Performance V1

The results indicate that logistic_v1 achieved the strongest overall performance in terms of ROC-AUC, Precision, and F1-score, while maintaining a balanced tradeoff between minority default detection and false positive generation. In contrast, xgboost_v1 obtained the highest Recall, demonstrating stronger sensitivity toward default detection at the expense of lower Precision.

Benchmark Performance Comparison – Version 2

The second experimentation stage incorporated additional hyperparameter adjustments and threshold optimization refinements in order to improve classification robustness and minority class detection performance.

Model Performance - V2				
Model Version	ROC-AUC	Precision	Recall	F1-Score
logistic_v2	0.7872	0.2374	0.3768	0.2913
random_forest_v2	0.7785	0.2035	0.4203	0.2742
xgboost_v2	0.7681	0.1718	0.5217	0.2585
lightgbm_v2	0.7633	0.183	0.4203	0.2549

Table IX .b. Model Performance V2

The updated experimentation results show marginal improvements for several models, particularly random_forest_v2, which improved both Precision and F1-score relative to their previous configuration. Nevertheless, logistic_v2 remained the strongest overall model, achieving the highest ROC-AUC, Precision, and F1-score among all evaluated configurations. Additionally, the benchmarking process revealed that ensemble boosting models such as XGBoost and LightGBM consistently achieved higher Recall values,

indicating stronger capability for identifying default observations. However, these gains were accompanied by lower Precision and reduced classification balance compared to the Logistic Regression configurations.

3.5 Performance Evaluation Results

Based on the benchmarking results obtained during the experimentation process, the two best-performing model configurations were selected for deeper evaluation and comparative analysis. The selection was primarily based on ROC-AUC, F1-score, classification balance, and minority default detection capability under the imbalanced credit risk setting.

According to the experimental results, `logistic_v2` and `random_forest_v2` demonstrated the strongest overall performance among the evaluated candidate models. Consequently, the following analyses focus on these two configurations in order to further evaluate discriminatory power, classification behavior, probability estimation quality, and business-oriented risk segmentation performance.

3.5.1 ROC - AUC

Training Set

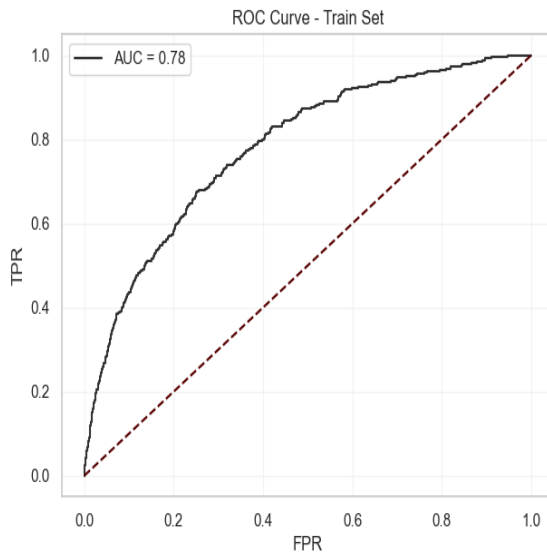


Figure XIII.a. ROC Curve Performance of *logistic_v2* - Train

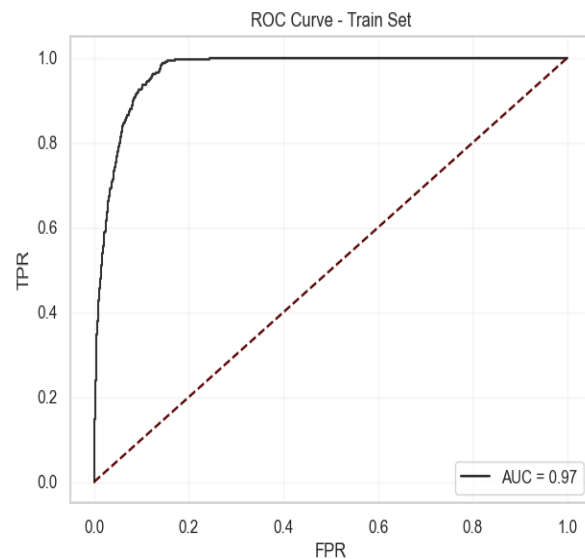


Figure XIII.b. ROC Curve Performance of *random_forest_v2* - Train

Test Set

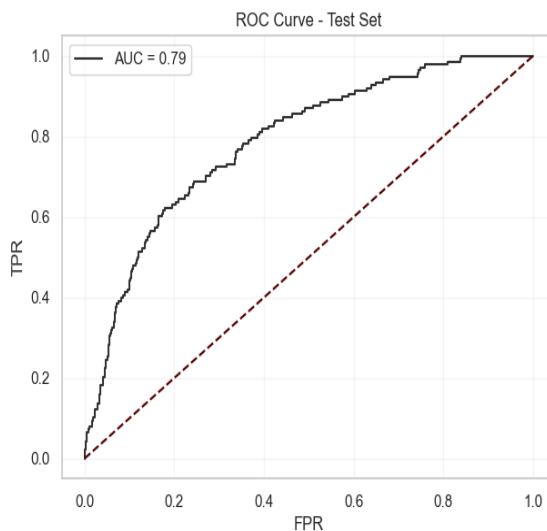


Figure XIII.c. ROC Performance of *logistic_v2* - Test

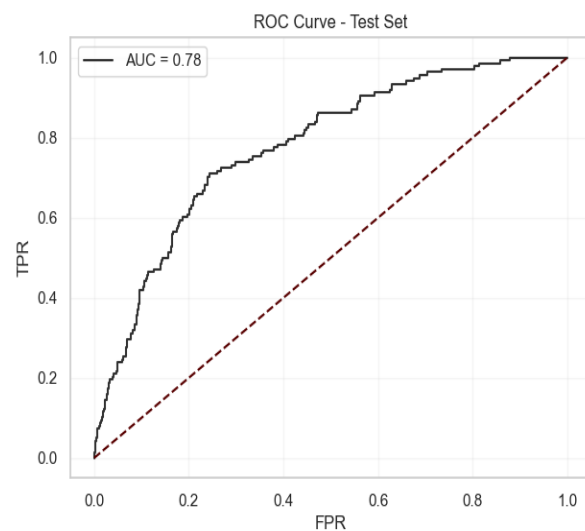


Figure XIII.d. ROC Performance of *random_forest_v2* - Test

The ROC-AUC results show that both models achieved strong discriminatory performance; however, *logistic_v2* demonstrated greater stability and generalization consistency between the training and testing datasets. Although *random_forest_v2* achieved near-perfect separation on the training set, its lower testing performance suggests mild overfitting behavior. In contrast, *logistic_v2* maintained a more balanced and robust predictive performance, supporting its suitability as the final deployment candidate within the proposed credit risk framework.

3.5.2 Confusion Matrix

Training Set

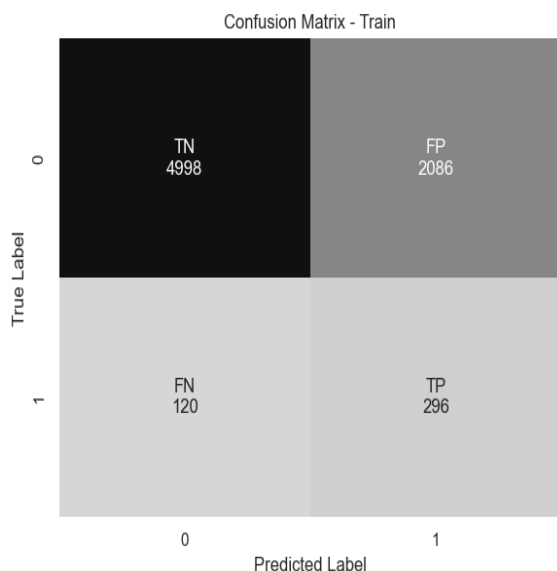


Figure XIV. a . Confusion Matrix of *logistic_y2* - Train

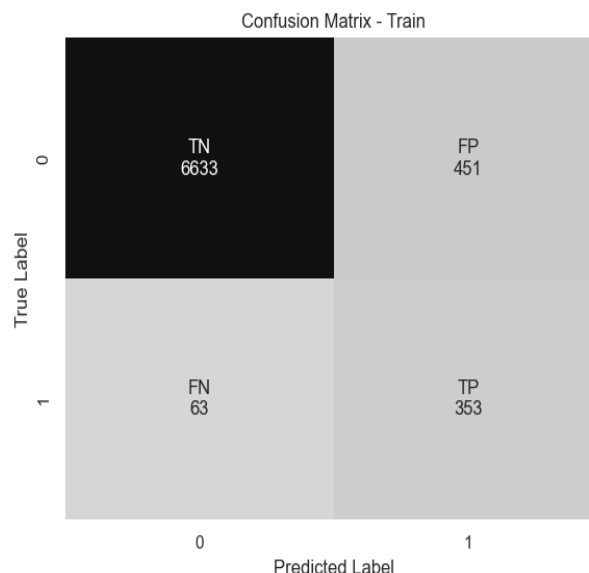


Figure XIV.b. Confusion Matrix - *random_forest_y2* - Train

Test Set

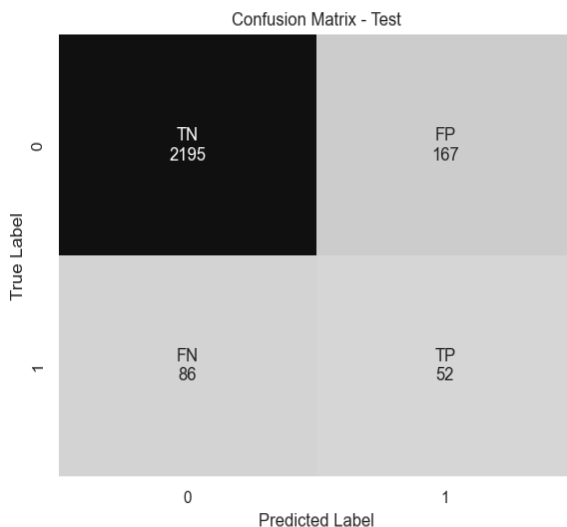


Figure XV. c . Confusion Matrix of *logistic_y2* - Test

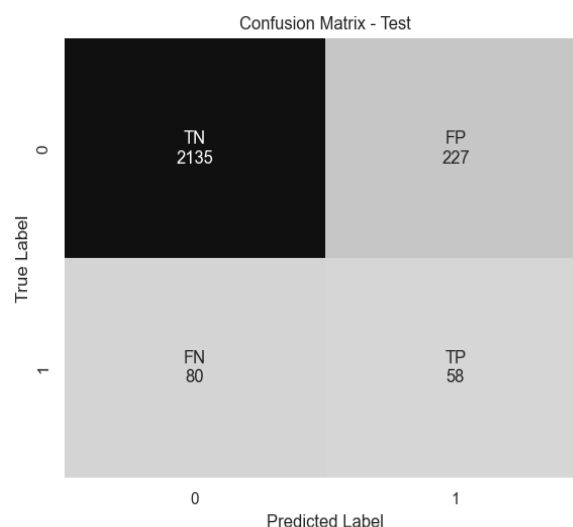


Figure XV.d. Confusion Matrix of *random_forest_y2* - Test

The confusion matrix results indicate that both models achieved strong classification performance in identifying non-default observations. However, *logistic_y2* maintained a more balanced tradeoff between false positives and false negatives, while *random_forest_y2* exhibited slightly higher sensitivity toward default detection at the cost of additional false positive classifications.

3.5.3 Predicted PD Distribution

Training Set

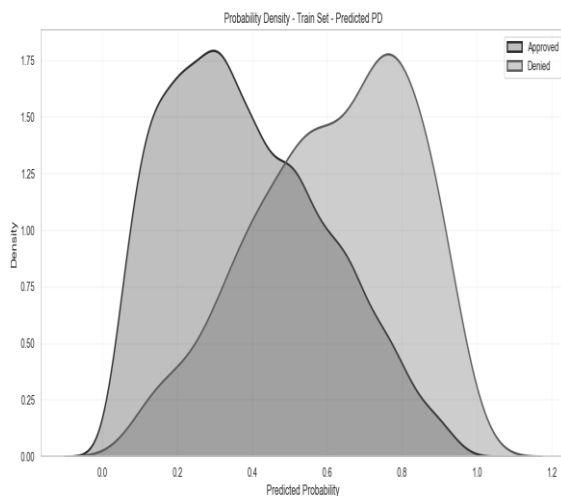


Figure XVI. a : Distribution of Predicted Probability of Default by Credit Decision Class of *logistic_v2* - Train

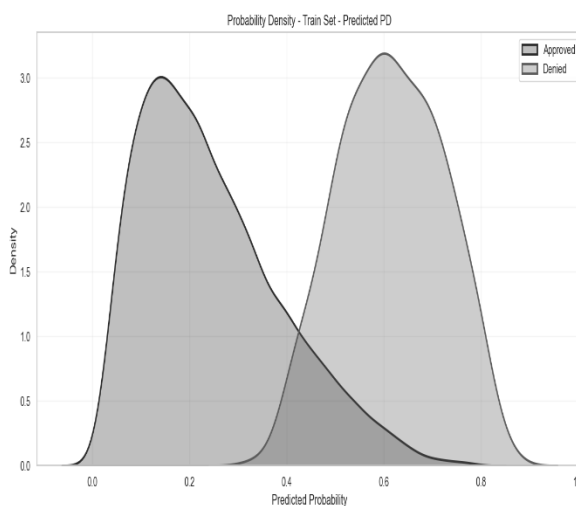


Figure XVI. b : Distribution of Predicted Probability of Default by Credit Decision Class of *random_forest_v2* - Train

Test Set

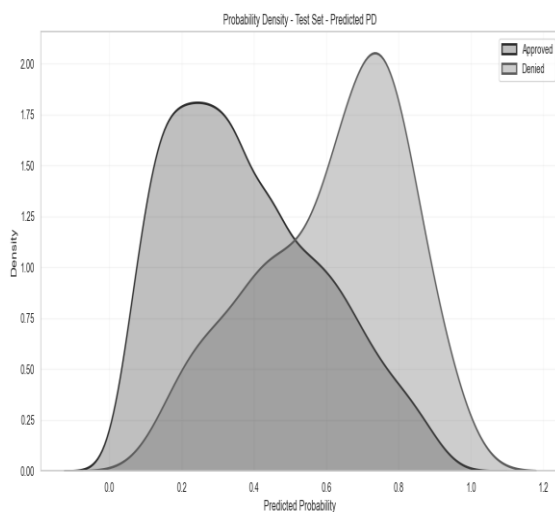


Figure XVII. a : Distribution of Predicted Probability of Default by Credit Decision Class of *logistic_v2* - Test

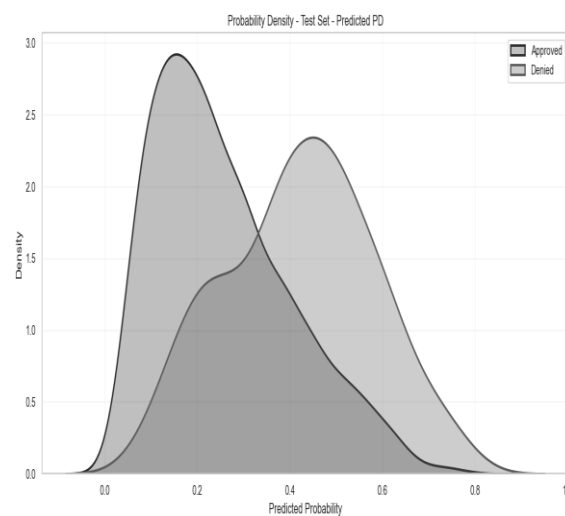


Figure XVII. b : Distribution of Predicted Probability of Default by Credit Decision Class of *random_forest_v2* - Test

The predicted PD distributions reveal distinct probability patterns between approved and denied observations for both candidate models. In particular, *logistic_v2* generated smoother and more interpretable probability segmentation, while *random_forest_v2* produced more concentrated probability regions with slightly greater overlap between classes, especially within the testing dataset.

3.5.4 Real vs Predicted PD Distributions by Class

Training Set

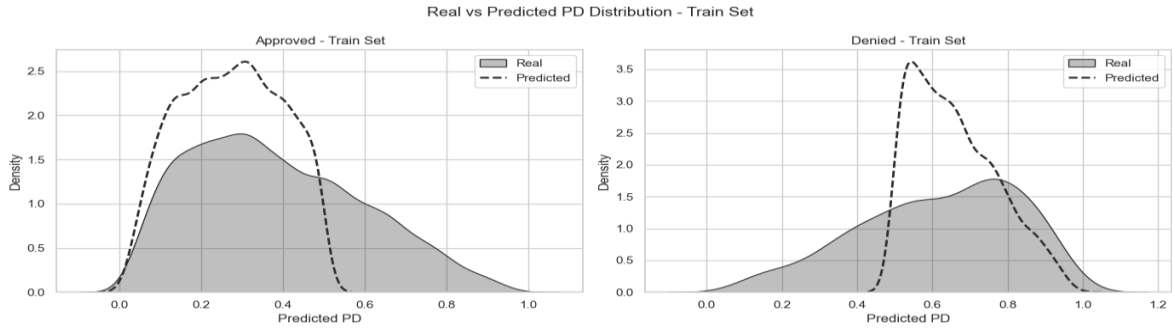


Figure XVIII. a: Comparison of Real and Predicted PD Distributions by Credit Decision Outcome of logistic_v2 - Train

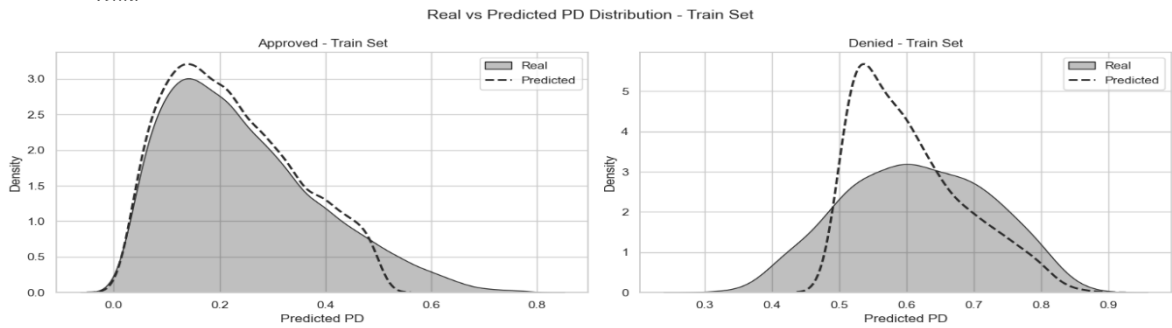


Figure XVIII. b: Comparison of Real and Predicted PD Distributions by Credit Decision Outcome of random_forest - Train

Test Set

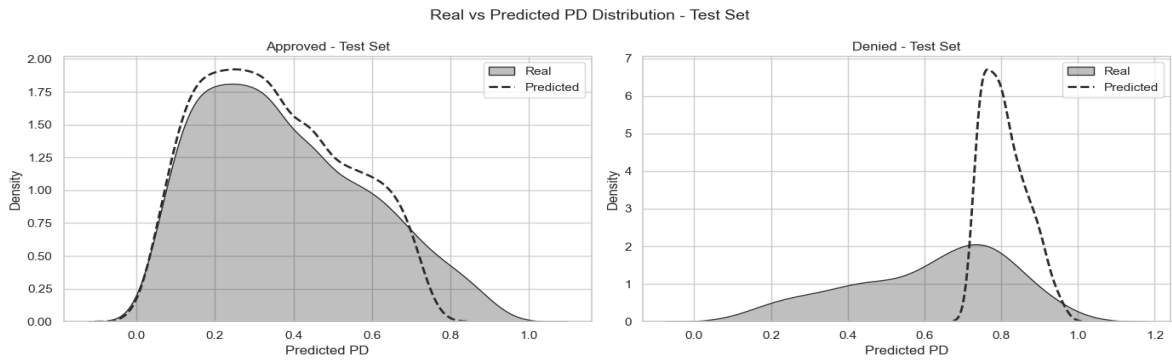


Figure XIX. a: Comparison of Real and Predicted PD Distributions by Credit Decision Outcome of logistic_v2 - Test

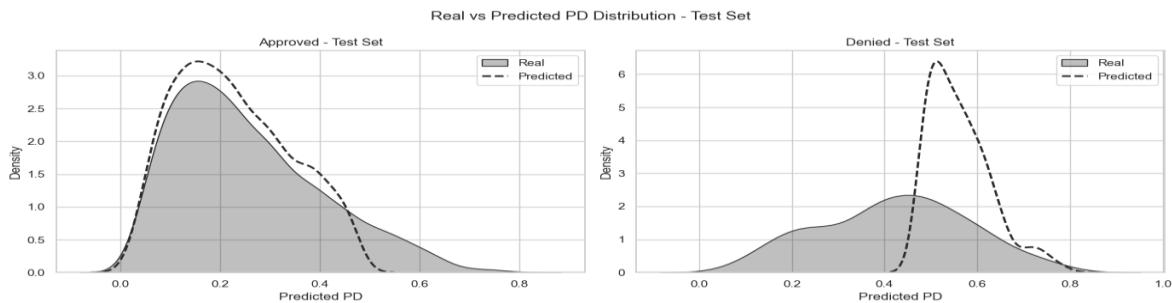


Figure XIX. b: Comparison of Real and Predicted PD Distributions by Credit Decision Outcome of random_forest_v2 - Test

3.5.5 Risk Bucket Analysis

Predicted Pd By Bucket

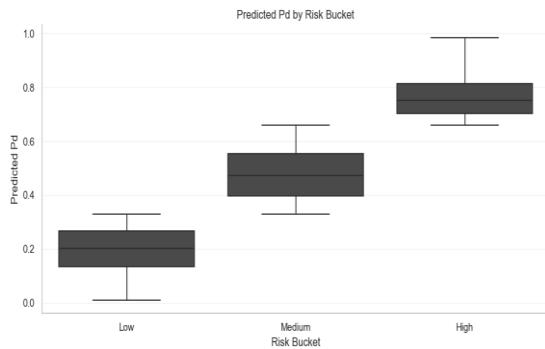


Figure XX. a : Distribution of Predicted Probability of Default Across Risk Buckets of logistic_v2

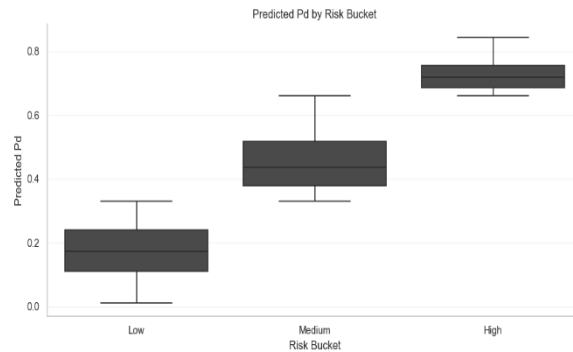


Figure XX. b : Distribution of Predicted Probability of Default Across Risk Buckets of random_forest_v2

Expected Loss By Bucket

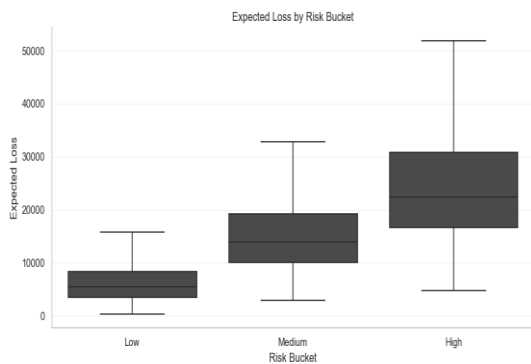


Figure XXI. a : Expected Loss Distribution Across Risk Buckets of logistic_v2

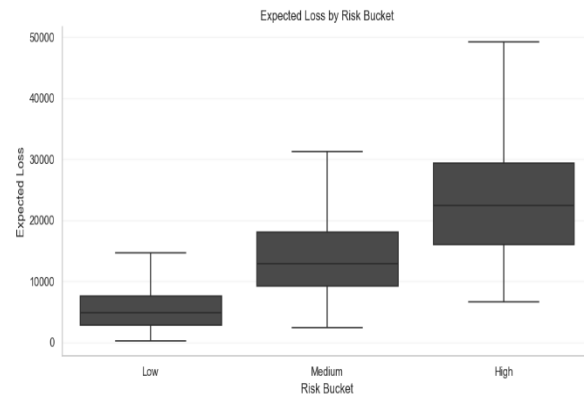


Figure XXI. b : Expected Loss Distribution Across Risk Buckets of random_forest_v2

Interest Rate By Bucket

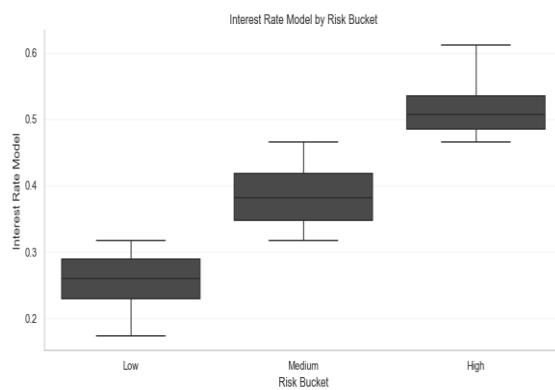


Figure XXII. a : Interest Rate Distribution Across Risk Buckets of logistic_v2

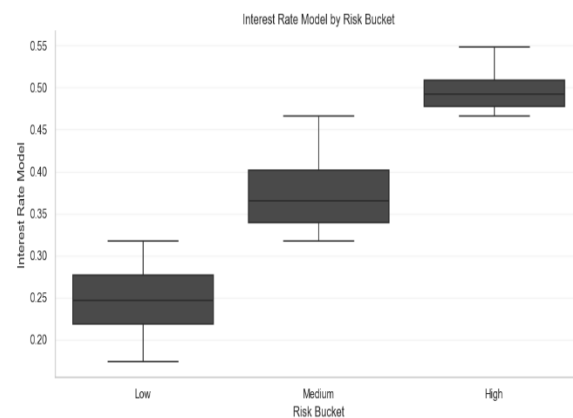


Figure XXII. b : Interest Rate Distribution Across Risk Buckets of random_forest_v2

The risk bucket analysis shows that both models were able to create a consistent progression of borrower risk across the Low, Medium, and High-Risk categories. As the assigned risk bucket increases, the predicted probability of default, expected loss, and interest rate levels also increase in a coherent manner, which indicates that the models are capturing financially meaningful risk patterns rather than arbitrary statistical groupings.

For the predicted PD distributions, both models clearly separate low-risk borrowers from higher-risk profiles, although *logistic_v2* exhibits smoother transitions and more stable segmentation across buckets. A similar pattern is observed in the expected loss analysis, where High Risk borrowers concentrate significantly larger loss exposures compared to Medium and Low Risk groups, reinforcing the economic consistency of the segmentation framework.

The interest rate distributions further validate the robustness of the proposed credit risk structure. Higher-risk borrowers systematically receive higher modeled interest rates, reflecting a realistic risk-adjusted pricing mechanism aligned with standard lending practices. In this context, the combination of PD estimation, expected loss behavior, and pricing segmentation demonstrates that the framework can support not only predictive classification tasks, but also practical credit decision processes such as borrower segmentation, pricing adjustment, and portfolio-level risk management.

3.6 Final Model Selection

Credit Risk Model Performance Metrics	
Metric	Value
Test ROC-AUC	0.79
Train ROC-AUC	0.78
Accuracy	89.9%
Precision	23.7%
Recall	37.7%
F1-Score	29.1%

Table IX. a: Final Model performance metrics

Following the benchmarking and evaluation stages, the candidate machine learning models were compared according to predictive performance, generalization stability, interpretability, calibration consistency, and deployment feasibility within the proposed

remittance-based credit risk framework. Although multiple models achieved competitive classification performance, the evaluation results indicate that *logistic_v2* provided the

strongest overall balance between discriminatory capability, stability across training and testing datasets, probability calibration consistency, and operational interpretability.

In particular, *logistic_v2* achieved the highest ROC-AUC and F1-score among the evaluated configurations while maintaining more stable behavior between training and testing environments relative to the ensemble-based alternatives. Additionally, the model generated clearer probability segmentation across borrower risk buckets and exhibited lower overfitting behavior compared to *random_forest_v2*.

From a business and deployment perspective, Logistic Regression also provides additional advantages due to its lower computational complexity, greater transparency, and easier interpretability for credit decision environments where explainability and regulatory consistency are highly relevant.

Consequently, *logistic_v2* was selected as the final deployment candidate for the proposed machine learning credit risk platform.

3.6.1 Final Hyperparameter Configuration

Logistic v2_ Hyperparameter Configuration		
Parameter	Value	Description
C	5.8442	Regularization strength controlling model complexity and coefficient penalization.
optimal_threshold	0.7244	Probability threshold selected through threshold optimization to maximize classification balance.

Table X : Final Model Hyperparameters Configuration

Table X summarizes the final hyperparameter configuration used for the selected *logistic_v2* model. The final configuration was selected through the Optuna optimization framework and

integrated within the MLflow experimentation pipeline to ensure reproducibility and consistent model tracking across experiments.

3.7 Credit rate construction

Since the product modeled for this project is a home improvement loan aimed at Mexican households receiving remittances in the state, without a formal mortgage on the property, the absence of this real guarantee is partially offset by direct debiting the remittance flow, similar to how Infonavit's Mejoravit program uses payroll deductions as an operational guarantee (Infonavit, 2025). This creates a mechanism that can reduce collection risk under normal

conditions, although it increases the Loss Given Default (LGD) compared to a traditional mortgage with real guarantees, given that the institution lacks a tangible asset to seize.

That said, the active interest rate for this project is determined using a risk-based pricing approach, which establishes that each client should pay a rate that accurately reflects their individual risk and the costs they generate for the institution (Kagan and Khartit, 2020; Hand and Henley, 1997). Under this idea, the rate charged to the client proposed in this project is broken down into six economic components, similar to what is presented by the institution "Bank for International Settlement" (2017).

$$Rate_{customer} = r_f + s_{anchorage} + (PD \cdot LGD) + c_{operat} + c_{capital} + m$$

where each term corresponds to a specific function within the credit operation, as described below.

- Component 1: Risk-free rate (r_f).

This corresponds to the minimum return that the institution must obtain on the borrowed funds, even in the absence of risk. In the Mexican financial system, the most commonly used benchmark for peso-denominated loan agreements is the 28-day TIIE (Interbank Equilibrium Interest Rate), published daily by the Bank of Mexico. As of May 8, 2026, the 28-day TIIE stood at 6.9971% (overnight Funding TIIE: 6.74%), which is the operational benchmark adopted for this exercise (Banxico, 2026a; Official Gazette of the Federation, 2026).

- Component 2: Funding spread ($s_{anchorage}$).

Given that the target segment (households without a credit bureau) is not served or considered by traditional commercial banks (which are funded with low cost demand deposits), and SOFOMs have demonstrated greater flexibility in operating with non-conventional profiles (Tu Hipoteca Fácil, 2021; Sofomes.com, 2025). Unlike commercial banks, our ENR SOFOM must issue securities, contract lines of credit with development banks (SHF, FIRA, NAFIN), or use its own capital, which increases its funding costs. Industry reports place the total funding cost for home improvement SOFOMs between 200 and 400 basis points (bps) above the 28-month TIIE (CNBV, 2024). That said, for this project, we assume a funding spread of

300 bps as the midpoint of the observed range, which leaves us with a total funding cost of 9.9971%.

- Component 3: Credit Risk Premium ($PD \cdot LGD$)

This component corresponds to the expected annual loss of the portfolio, expressed as a percentage of the outstanding balance. The probability of default (PD) comes directly from the model trained on the project, with an AUC of approximately “0.7872”. The Loss Given Default (LGD) was calibrated at 45%, an intermediate value between the secured mortgage range (20%–35%) and unsecured consumer credit (close to 84% in major Mexican banks). This was based on the fact that, although the product does not constitute a formal mortgage, direct debit collection from the receiving account reduces the operational risk of collection (Investopedia, 2024; Moody's Local Mexico, 2025; Office of the Comptroller of the Currency, 2007). The resulting risk premium by segment is:

Group	Mean PD	LGD	Premium ($PD \cdot LGD$)
Low Risk	0.1981	0.45	0.0891
Medium Risk	0.4782	0.45	0.2152
High Risk	0.7644	0.45	0.3439

Table XI : Risk Premium Estimation by Borrower Risk Segment

- Component 4: Unit operating costs (C_{operat})

These are broken down into four categories according to the financial statements of comparable institutions (CNBV, 2024; HR Ratings, 2026):

Category	Cost (%)
Origination (annualized)	0.80%
Monthly administration	2.00%
Direct debit collection	0.30%
AML/CFT Compliance	0.40%
TOTAL	3.50%

Table XII : Operational Cost Structure for Interest Rate

The collection component is reduced because the collection is done by automatic debit on the remittance receiving account, eliminating telephone or face-to-face management under normal conditions (CONDUSEF, 2018).

Component 5 — Cost of Regulatory Capital ($c_{capital}$).

To ensure the project is as realistic as possible, it was decided to incorporate the regulations derived from Basel III, which require Mexican multiple banks to maintain a minimum regulatory capital equivalent to 10.5% of risk-weighted assets, comprised of core capital 1 (7%), conservation capital (2.5%), and other components (CNBV, 2018; Banxico, 2022). While not strictly subject to the Basel regulatory framework, SOFOMES ENR (non-bank financial institutions) adopt similar standards in practice to access development banking lines of credit and issue debt with reasonable ratings. For this product, the risk weighting is 100% (loan without formal mortgage guarantee, according to the Capitalization Rules; Mexican Banking Association, 2018). The cost of capital is calculated as:

$$c_{capital} = w_{risk} \cdot k_{min} \cdot ROE_{obj} = 1.0 \times 0.105 \times 0.18 = 1.89\%$$

donde el ROE objetivo del 18% es representativo de un negocio nuevo de inclusión financiera con perfil de riesgo medio-alto, dentro del rango histórico de la banca comercial mexicana (entre 14% y 18%) y del rango de SOFOMEs especializadas (entre 18% y 22%) reportado por Banxico (2025).

- Component 6: Operating Profit Margin (m).

This last concept is used to capture the unexpected risk of the portfolio (variance of the loss beyond its expected value), the opportunity cost of voluntary economic capital, and the net profit required to sustain business growth. For this study, a margin of 150 basis points (1.50%) is adopted, above the regulatory cost of capital, justifying the novelty of the product with increased uncertainty in the segment served and also with the need to generate additional voluntary reserves during the first years of operation. The net financial margin of the housing portfolio in Mexico has historically ranged between 4% and 6%, according to the quarterly reports of the Bank of Mexico (Banxico, 2025).

- Integration

Finally, since five of the six components are common to all buckets (independent of the specific applicant), they are integrated together by summing them, while the credit risk premium varies according to the average PD of the bucket in which the model classifies each applicant. The result of the independent parameters ultimately yields 16.8871%. But this does not include the final risk premium assigned to each customer's segment.

Group	Premium (PD · LGD)	All components	Premium (PD · LGD)
Low Risk	0.0891	0.168871	28.25% ±3%
Medium Risk	0.2152	0.168871	37.70% ±3%
High Risk	0.3439	0.168871	47.90% ±3%

Table XIII : Estimated Risk-Adjusted Interest Rate Ranges by Borrower Segment

3.8 Model Validation

3.8.1 Validation techniques and results obtained

For the validation of this model, two approaches were used to compare its performance and applicability. The first is the statistical validation of the predictive model, which evaluates the algorithm's discriminatory capacity to separate households with and without default within the dataset. The second is from the perspective of a credit examiner, through an economic validation of the results: confirming that the estimated default probability and resulting interest rates are consistent with the market environment in which the institution would operate, specifically in the state of Jalisco.

The statistical validation strategy for model evaluation was carried out by dividing the dataset into a training set (75%) and a test set (25%), preserving the proportion of the positive class (households in default, 5.5% of the total) in both partitions. This decision is critical because a model that always predicted the majority class would achieve an accuracy of 94.5% without any real predictive value. Therefore, conventional accuracy metrics can lead to a misleading assessment of performance. Therefore, the use of robust metrics such as ROC-AUC, PR-AUC, F1-Score and confusion matrix inspection is often recommended (Thomas, 2000; World Bank Group, 2019).

3.8.2 Analysis of the model's accuracy and effectiveness

“ The validity of a credit scoring model is not limited to its predictive power in historical data, but to its ability to generate credit decisions that are financially sustainable in the real market environment (Hand and Henley, 1997) ”

As proof of the coherence of this project in the industrial sector, the interest rates derived from the model are compared with the real landscape of housing finance in Jalisco and Mexico, verifying that the prices assigned to each risk segment are sufficient to cover the expected costs and losses, and that they are also relevant and valid to be a product for unbanked segments.

As seen in the results presented above, the final rates obtained are significantly higher than those of conventional commercial banks, as they respond to a financial logic that differentiates them from traditional mortgage loans.

The first point for any validation of housing loan rates in Mexico is the average Annual Percentage Rate (APR) of the mortgage market, reported by the Bank of Mexico and CONDUSEF (National Commission for the Protection and Defense of Financial Services Users). At the beginning of 2025, the standard mortgage loan in Mexico had an average APR of 14.13%, with a minimum of 11.20% and a maximum of 28.18%. However, this range corresponds to products offered by commercial banks and public agencies such as Infonavit and Fovissste to formal workers with verifiable credit history, verifiable payroll income, and real property collateral. However, remittance-based loans for Jalisco present three important differences; The first issue is that the proposed product operates without a real mortgage guarantee, which translates into a calibrated LGD of 45%, reflecting a loss level in the event of default significantly higher than the 14.5%-20% observed in traditional mortgage portfolios in Mexico. Furthermore, the interest rates and APRs for mortgage loans offered by SOFOMs are higher than those of mortgage loans granted by banks. Second, the product serves a segment without a formal credit history, which implies higher evaluation costs and a greater risk of adverse selection. The 2024 National Survey of Financial Inclusion (ENIF) documents that financial exclusion persists, especially in rural and semi-urban areas with a high concentration of remittance-receiving households, precisely the project's target market (INEGI-CNBV, 2024).

And third, the funding operates under a SOFOM ENR scheme, whose deposit costs are structurally higher than those of commercial banks, as documented in the rate construction section of this report.

3.8.3 Comparison with the unsecured personal loan segment

For a realistic comparison of this product to unsecured personal and home improvement loans offered by SOFOMES (non-bank financial institutions) and microfinance institutions, the average interest rate weighted by balance for personal loans between 5,000 and 30,000 pesos was 59.3 percent, according to the Bank of Mexico's Basic Indicators of Personal Loans, with data as of February 2024. Eighty percent of the outstanding balance of personal loans was concentrated in rates between 23.0 and 76.0 percent, with half of the balance granted at rates less than or equal to the median rate. Now, for longer-term personal loans (more than 36 months), the data from the same report indicates that the average weighted rate is around 30.2 percent, although this segment represents only 7.5 percent of the number of loans but accounts for 45.1 percent of the outstanding balance (Banxico, 2023; Banxico, 2024).

That said, a Remittance Loan like the one presented is, in terms of its risk structure, closer to a medium-term personal loan without collateral than to a conventional mortgage; but with the peculiarity that the direct debit collection on the remittance receiving account in this case acts as a mechanism for partially mitigating the collection risk, without constituting an enforceable real guarantee.

In this context, Jalisco, as the third largest recipient of remittances nationwide, has a significant proportion of households in rural municipalities where the alternative financing these families rely on comes not from traditional banks but from informal sources. Informal lending institutions and informal lenders in Mexico offer implicit interest rates of 100% to 300%+ annually. SOFOMES (non-bank financial institutions) and fintechs that serve individuals without formal credit histories operate with Annual Percentage Rates (APRs) of 30% to 120%+ annually, depending on the borrower's profile. Therefore, the rates offered by this model (28.25%-47.90%) represent a financial inclusion option with a significantly lower cost than the benchmark informal financing for this segment.

3.8.4 Financial Model: Portfolio performance and model performance

As part of the appendices to this project, the financial model's financial projections show a weighted average portfolio rate of 34.84%, resulting from the projected origination composition (40% Low Risk, 50% Medium Risk, 10% High Risk) and the three rates per segment. This net interest margin (NIM) of 25.9% on the average portfolio is high compared to Mexican commercial banks (5%-7%), but consistent with the risk profile of SOFOMES (non-bank financial institutions), whose reported NIMs range between 8% and 15% according to HR Ratings (2026), and may exceed this range in institutions with high financing costs and higher-risk segments.

Furthermore, the sensitivity analysis of the financial model confirms the robustness of the interest rate structure; given a weighted portfolio differential (PD) between 3% and 9% and a long-term debt service (LGD) between 25% and 65%, the Year 3 ROE remains positive in virtually all scenarios except the most stressful ones (high PD and high LGD simultaneously). The projected indicators at the end of Year 3 are as follows:

Indicator	Year 1	Year 2	Year 3	Reference
ROE	5.40%	0.803	75.20%	Target: 18% (multiple banks MX: 15-22%)
ROA	1.30%	0.101	11.40%	SOFOMES MX: 0.5-4.5% (HR Ratings)
IMOR	3.00%	0.03	3.00%	CNBV Housing 2024: 3%
ICAP	11.50%	0.126	17.00%	Basel III minimum: 10.5% (CNBV, 2018)
Coverage ratio (Reserves / CV)	1.30x	1.30x	1.30x	SOFOMES MX: 0.8x-2.1x (HR Ratings)

Table XIV : Financial Sustainability and Portfolio Risk Indicators Across the Projection Horizon

3.9 Practical Application

3.9.1 Specific industry use case.

Housing market in Jalisco

The specific case for the model is the origination of home improvement loans for remittance-receiving households in the 124 municipalities of the state of Jalisco, with an emphasis on the medium- and high-migration-intensity municipalities identified in Component 1 of the project. Jalisco presents ideal conditions for this product: in 2024, loans for home improvements are projected to total 18.837 billion pesos nationwide, representing a 32.3% increase compared to the amount granted in 2014. Despite this growth, the supply is concentrated in government programs (Infonavit, Mejoravit) that require formal employment, excluding remittance-receiving households in the informal sector.

The model proposed in this case functions as a three-stage system for the origination process of the SOFOM ENR (Non-Bank Financial Institution):

Admission

The applicant must have received electronic remittance transfers for at least 24 months in the bank account provided. This condition is digitally verifiable and eliminates the need for proof of income from payroll.

Scoring and Classification

The model calculates the household's credit score based on the 16 variables selected in the characteristic analysis, SHAP, and variable selection stages. The score is transformed to the FICO 300-850 scale and assigns the applicant to one of three risk buckets.

Sign terms and Conditions

The operational policy for rates and rejection, derived directly from the risk-based pricing system, automatically determines the loan conditions:

Group	Final Rate	Maximum amount (MXN)	Maximum term	Decision
Low Risk	28.25%	500000	15 years	Automatic approval
Medium Risk	37.70%	300000	10 years	Approval with verification
High Risk	47.90%	100000	5 years	Conditional approval or rejection

Table XV : Proposed Credit Policy Conditions by Borrower Risk Segment

For this model, the average projected loan amount is \$200,000 MXN, within the \$50,000-\$500,000 MXN range defined in the product's assumptions. This amount was chosen to align with the typical cost of home improvement projects in Jalisco municipalities with high migration rates: room additions, roofing, and plumbing and electrical installations, which represent the most common use of remittance income for housing according to BBVA Research (2025).

3.10 Simulation and results analysis.

3.10.1 Financial projections for 3 years

To simulate the project's implementation in a "realistic" scenario, a financial projection model was built (file MC_ProyectoProyecciones_C_Remesas.xlsx) to simulate the operation of the SOFOM ENR during the first three years under the following origination assumptions: 500 loans in Year 1, with 30% growth in Year 2 (650 loans) and Year 3 (845 loans), with an average loan amount of \$200,000 MXN. The portfolio mix assumes 40% Low Risk, 50% Medium Risk, and 10% High Risk, consistent with the historical distribution of the synthetic portfolio.

For this exercise, the main projected results are:

Portfolio growth

The gross portfolio grows from \$92 million at the end of Year 1 to \$320.9 million at the end of Year 3. The average annual balance (the basis for calculating interest income) increases from \$46 million in Year 1 to \$258.9 million in Year 3.

Financial income

Interest income grew from \$16.0 million in Year 1 to \$90.2 million in Year 3, reflecting the combined effect of portfolio growth and the weighted average annual rate (34.84%). Additionally, origination fees (1% of the loan amount) generate \$1.0 million, \$1.3 million, and \$1.7 million in each year, respectively.

Portfolio quality

The projected non-performing loan ratio (NPL ratio) remains stable at 3.0% over the three-year period, aligned with the mortgage portfolio delinquency rate reported by the CNBV (2024) for the housing segment. Likewise, the coverage ratio (loan loss reserves) stands at 1.30x, within the healthy range of 0.8x-2.1x reported by HR Ratings (2026) for SOFOMES (non-bank financial institutions).

Profitability

Net income increases from \$558,888 in Year 1 (the startup year with high initial investment) to \$14.2 million in Year 2 and \$29.8 million in Year 3. ROE jumps to 80.3% in Year 2 and stabilizes at 75.2% in Year 3, levels well above the target of 18%. This is due to the high operating leverage of a rapidly expanding business where the initial capital of \$10 million remains constant while the portfolio grows.

Capital recovery

The model projects a return on initial capital of approximately 1.67 years, indicating that the initial investment is recovered during the second year of operation. Finally, given that the project's NPV is positive and the IRR of equity exceeds the cost of capital (K_e), the project's viability as an investment could be considered a priori.

	Año 0	Año 1	Año 2	Año 3	Total / VPN
1. Flujo de Caja Libre de la Firma (FCFF) (antes de financiamiento)					
Utilidad de operación (UAFIR antes de impuestos)	-	\$887,124	\$22,470,006	\$47,236,529	
(+) Gasto financiero (revertir, ya está en UAFIR)	-	\$4,104,366	\$12,887,710	\$23,097,731	
EBIT (resultado de operación antes de intereses e impuesto:	-	\$4,991,491	\$35,357,715	\$70,334,260	
(-) Impuestos teóricos sobre EBIT (ISR efectivo)	-	(\$1,846,852)	(\$13,082,355)	(\$26,023,676)	
NOPAT (utilidad operativa neta después de impuestos)	-	\$3,144,639	\$22,275,361	\$44,310,584	
(+) Depreciación y EPRC (gastos no efectivos)	-	\$4,835,729	\$8,062,732	\$11,906,629	
(-) Capital de trabajo	-	(\$88,054,266)	(\$100,739,507)	(\$119,075,615)	
(-) Inversión en activo fijo (CapEx)	-	(\$250,000)	-	-	
FCFF (Free Cash Flow to Firm)	-	(\$80,323,897)	(\$70,401,414)	(\$62,858,402)	
2. Flujo de Caja Libre al Accionista (FCFE) (para los socios)					
FCFF (línea anterior)	-	(\$80,323,897)	(\$70,401,414)	(\$62,858,402)	
(-) Intereses pagados × (1 - tasa efectiva impuestos)	-	(\$2,585,751)	(\$8,119,257)	(\$14,551,570)	
(+) Variación neta en pasivo financiero (deuda)	-	\$78,200,000	\$89,148,000	\$105,382,320	
FCFE (Free Cash Flow to Equity)	-	(\$4,709,648)	\$10,627,329	\$27,972,347	
3. Flujo del inversionista (incluye inversión inicial y valor terminal)					
Aportación de capital (Año 0)	(\$10,000,000)	-	-	-	
FCFE operativo (línea anterior)	-	(\$4,709,648)	\$10,627,329	\$27,972,347	
(+) Valor terminal en Año 3 (perpetuidad Gordon)	-	-	-	-	
Flujo total del inversionista	(\$10,000,000)	(\$4,709,648)	\$10,627,329	\$27,972,347	
4. Descuento de flujos y valoración del proyecto					
Periodo (t)	-	1	2	3	
Factor de descuento $[1 / (1 + Ke)^t]$	1.0000	0.7407	0.5487	0.4064	
Valor presente del flujo del inversionista	(\$10,000,000)	(\$3,488,628)	\$5,831,182	\$11,369,140	\$3,711,694
5. Indicadores de valoración del proyecto					
VPN del proyecto (suma de VPNs)		\$3,711,694			
TIR del equity (Internal Rate of Return)		49.39%			
Múltiplo del capital invertido (MOIC = Σ FCFE / Aportación)		3.39x			
Valoración por múltiplo P/E × UN Año 3 (cross-check)		\$148,795,067			
VP del valor por múltiplo de salida (descontado a hoy)		\$60,476,581			

Table XVI : Model Cash Flow

3.11 Impact on existing processes and problem-solving.

The model was designed to generate operational, social, and financial value in three main areas:

Automation of the credit evaluation process, since the credit scoring system replaces manual application review for 7.23% of the potential borrower pool (Low Risk), eliminating the bottleneck of subjective analysis and reducing response time. For Medium Risk (85.66% of the pool), verification is limited to documentary confirmation of the remittance account, without the need for extensive credit investigation. Only the High-Risk segment (7.11%) requires an additional conditional evaluation. This workflow substantially reduces the unit origination cost compared to a completely manual process, estimated in the model's assumptions at 80 basis points annualized on the outstanding balance.

Financial inclusion with fair pricing, which perhaps is the second most valuable contribution, and that is that the product directly addresses the problem documented by CEMLA (2025)

and BBVA Research (2025), in which, despite having regular and verifiable income streams, many households are excluded from the formal credit system. Therefore, by assigning risk-based rates (instead of rejecting all applicants without a credit history), the model maximizes inclusion without compromising the institution's financial sustainability. A household with regular remittances for five years, low cash flow volatility, and homeownership in a municipality with a high Regional Score can access home improvement loans at annual rates of 28.25%, substantially lower than the implicit rates of informal financing (100%-300% annually).

Formalizing remittance flows as alternative collateral, which is presented as the area with the greatest impact, as the model establishes a replicable precedent for the Mexican financial system, demonstrating that the temporal pattern of electronic remittance transfers (median, variability, trend, interruptions) can replace credit bureau history as a predictor of ability and willingness to pay. This is consistent with the World Bank Group's guidelines (2019) on the use of alternative data in credit scoring for unbanked populations, and with the CONDUSEF financial inclusion agenda (2024).

3.12 Deployment and ML System Integration

Following the model development and validation stages, the proposed credit risk framework was integrated into a web-based analytical platform designed to operationalize the machine learning outputs and provide an interactive visualization environment for municipality-level credit risk analysis. The deployment phase focused on transforming the predictive framework into a scalable and interpretable analytical system capable of integrating geospatial visualization, machine learning monitoring, and borrower risk segmentation within a unified interface.

The platform combines the outputs generated by the final machine learning pipeline with geospatial municipality information and financial risk metrics, allowing users to interactively explore modeled credit risk behavior across the state of Jalisco.

3.12.1 Interactive Geospatial Dashboard

The first component of the deployed platform consists of an interactive geospatial dashboard developed to visualize municipality-level risk segmentation across Jalisco. The application integrates the outputs produced by the final machine learning model, including predicted probability of default (PD), expected loss estimations, approval rates, and borrower risk bucket classifications.

The dashboard was developed using a frontend architecture based on HTML and CSS for the user interface and visualization layer, while the backend processing and machine learning integration were implemented in Python. The application was deployed through the Vercel cloud platform, allowing the system to be accessed directly through GitHub integration or via a public deployment link.

The geospatial layer was constructed using municipality polygon JSON files containing regional boundary information. This structure allows the framework to be scalable and adaptable for additional states across Mexico. Consequently, the implementation can be extended to other regions of the country by incorporating the corresponding municipality-level JSON geographic files and integrating the associated regional financial indicators into the processing pipeline.

The dashboard provides an operational visualization environment capable of displaying:

- Municipality-level predicted PD
- Risk bucket classification
- Expected loss estimation
- Approval rate metrics
- Geographic borrower segmentation
- Regional credit risk concentration patterns

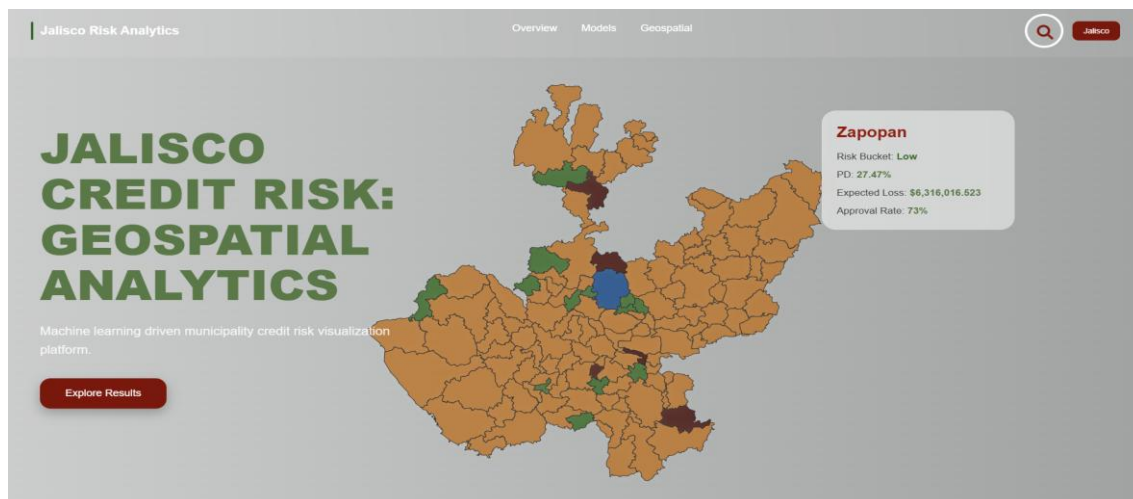


Figure XXIV : Interactive Geospatial Dashboard for Municipality-Level Credit Risk Visualization in Jalisco

The platform transforms the predictive framework into a practical analytical tool capable of supporting regional credit monitoring and geospatial borrower segmentation within remittance-based lending environments.

3.12.2 Machine Learning Monitoring and Analytics Dashboard

In addition to the geospatial visualization layer, the platform also incorporates an integrated machine learning analytics interface for monitoring model performance and evaluation metrics generated during the experimentation process.

This analytical module allows the visualization of training and testing metrics associated with the final selected model, including ROC-AUC behavior, confusion matrices, probability distributions, and risk segmentation analytics. The integration of these components enables continuous monitoring of model stability and predictive behavior within the deployed environment. The analytics interface was designed to centralize the outputs generated through the MLflow experimentation framework and the evaluation pipeline, providing a unified environment for model inspection and performance validation.

Figure XXV presents the machine learning monitoring and analytics interface integrated within the deployed web platform.

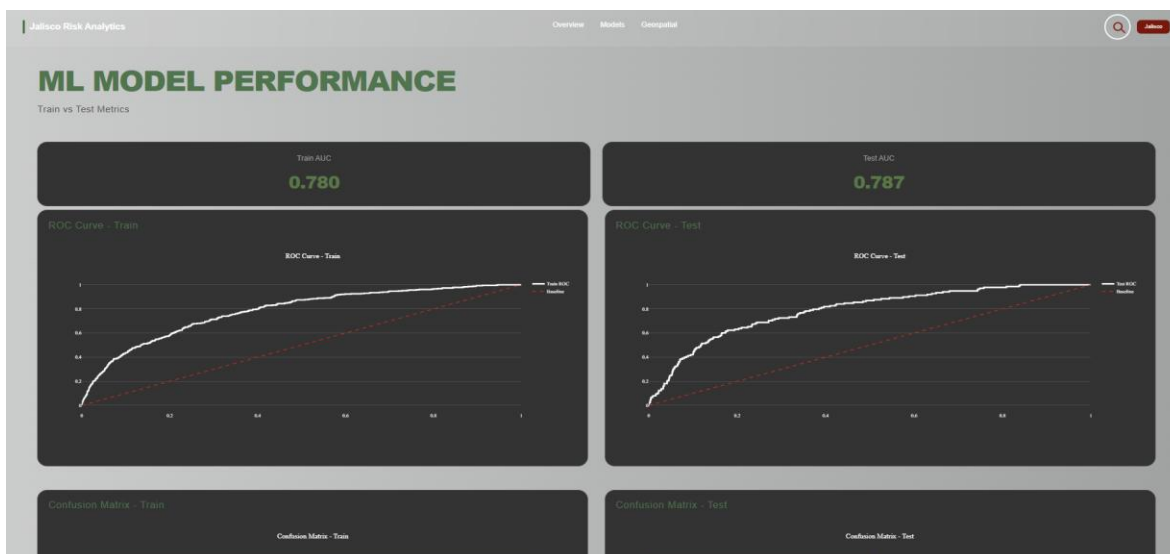


Figure XXV : Machine Learning Monitoring and Performance Analytics Dashboard

The integration of predictive modeling, evaluation analytics, and geospatial visualization within a single deployed platform demonstrates the transition of the project from an isolated machine learning model into an operational end-to-end analytical credit risk system.

4. Impact Assessment

Estimation of time saved, increase in user conversion, profitability, and/or performance.

Based on the financial model's indicators within the Excel spreadsheet, it is estimated that for origination efficiency, the model projects a Customer Acquisition Cost (CAC) that decreases as a proportion of interest income in the first year, from 4.31% in Year 1 to 3.29% in Year 3, reflecting economies of scale in the digital origination process. This is reinforced by the portfolio productivity per employee, which increases from \$11.5 million in Year 1 to \$24.7 million in Year 3.

Likewise, regarding applicant conversion, the group approval policy allowed 92.89% of the potential applicant pool (mostly comprised of Low and Medium Risk) to become clients. Under a traditional system, these applicants would likely have been rejected solely due to the lack of a formal credit history. Only 7.11% of the pool (High Risk) remains subject to restrictive conditions or final rejection.

Finally, regarding the profitability of the project, the accumulated net profit at the end of the three years was \$44.5 million on an initial capital investment of \$10 million, with a payback of 1.67 years and a stabilized ROE above the target of 18% from Year 2.

5. Triple impact assessment (social, environmental, and financial).

To have a financial and social impact, the proposed Remittance Credit project aims to directly reduce the housing finance access gap for remittance-receiving families in Jalisco, a segment that, according to CEMLA (2025), represents one in every 3.1 families in communities with fewer than 2,500 inhabitants in the Western Bajío region. The possibility of accessing home improvement loans with terms of 8-15 years allows them to complete housing projects more quickly, eliminating the inefficient process of saving money "under the mattress" for years, which prevents families from having opportunities to grow. In this sense, home improvement can have many positive effects on children's health, school performance, and family well-being (World Bank Group, 2019). Additionally, due to the very nature of this model, each customer using the product will be progressively formalizing their finances simultaneously, as households that make timely payments on their loans are already building a credit history, which will improve their future access to other formal financial products at lower rates.

In line with the sustainable urban development goals of the state of Jalisco and the National Housing Program, the presented product, whose main purpose is home improvement, has the impact of reducing pressure on agricultural and peri-urban land. This provides an opportunity to reuse and make better use of existing housing instead of abandoning it and building a new one; thus minimizing the generation of demolition waste and reducing the carbon footprint associated with the mass production and use of construction materials. In Jalisco municipalities with high migration rates, where unfinished housing (unfinished concrete blocks, sheet metal roofs) is common, the home improvement loan contributes to housing densification without urban sprawl (SEDATU, 2024).

6. Business Model

The financial projection model assumes a static operating environment over the three-year horizon, where average loan amounts, origination growth rates, and other factors remain relatively unchanged. In practice, this composition will vary as the institution gains experience. Furthermore, in many cases, origination growth will depend heavily on factors outside the model, such as the pace of regulatory registration with CONDUSEF, the speed of establishing credit lines with development banks like SHF or NAFIN, and the institution's ability to expand to areas where the target population is concentrated.

A second limitation lies in the assumed financing cost; this assumes a fixed financing spread of 300 basis points above the 28-day TIIE (Interbank Equilibrium Interest Rate) throughout the projection period. This was established as a midpoint of the range observed for SOFOMs (Multiple Purpose Financial Institutions) specializing in home improvement (200-400 bps according to CNBV, 2024). However, in reality, a newly inaugurated or recently established ENR SOFOM (Non-Bank Financial Institution) without a credit rating would likely face spreads exceeding this range during its initial years. The sensitivity analysis in the financial model addresses this issue somewhat by stress-testing the net present value and net income for Year 3 at financing spreads between 200 and 450 basis points, confirming that the project remains viable within this range. However, if the TIIE continues its downward trajectory, the fixed-rate nature of the product could reduce this interest rate differential over time, which a future version of the model should address through a variable-rate clause or rate adjustment linked to TIIE fluctuations.

As documented in the introduction, Mexico experienced its first annual decline in remittances since 2013, due to the tightening of immigration control measures in the United States and the proposed federal tax on remittances sent abroad. Jalisco's consumption, close to 10%, was higher than the national average (Lider Empresarial, 2026). A reduction in remittance flows would diminish the quality of the guarantees backing repayment. This systemic risk is not modeled as such in the financial projections and represents perhaps the most significant assumption underlying the product's viability.

Finally, the business model assumes a 100% digital and remote operating model. This is an efficiency assumption that may not hold true in practice for municipalities with low digital presence. According to the 2024 National Survey of Financial Inclusion (INEGI-CNBV, 2024), the use of digital financial services remains significantly lower in rural areas and among older age groups, both of which are included in the target population.

6.1 Data Construction

This is the most limited part of the project. First, the registry of 10,000 households was constructed by calibrating parametric distributions with official aggregate statistics from Banxico, CONAPO, INEGI, CEMLA, BBVA Research, and CNBV, instead of using actual records of individual transactions. The reason for this decision was that individual remittance histories are not publicly available due to privacy concerns and internal institutional policies. This means that the model was trained with simulated behavior, designed to be statistically consistent with observed aggregate patterns, but could not be directly validated with actual household-level credit outcomes. This distinction is important for two reasons. First, the default rate calibration is based on the mortgage portfolio delinquency rate reported by the CNBV (2024), adjusted upward to reflect the higher-risk profile of the target segment. While this could be a reasonable approximation, the actual default rate is unknown because no comparable product has operated on a large scale in Jalisco. Second, the trained model has not been exposed to the noise that real-world transaction data would contain, such as irregular transfer schedules, exchange rate volatility, seasonal distortions from specific senders, and extreme situations affecting the applicant's personal circumstances, such as illness, job loss, or family changes on the sender's side.

On the other hand, the temporal disaggregation process used to convert quarterly municipal remittance data to monthly frequency (Chow-Lin method) introduces an additional layer of uncertainty into the estimate. The disaggregated series is an optimal unbiased linear estimate conditioned on the national monthly series as an indicator, but it is not based on observed data. While the validation of the aggregation constraint yielded a virtually zero error at the quarterly level, and a seasonal correlation of $r = 0.9842$ between the disaggregated series for Jalisco and the national indicator, in reality, in municipalities with low volumes, the

disaggregated monthly series may exhibit an estimation variance that the aggregate validation does not detect.

One improvement to consider for the future would obviously be to use more realistic, predictive data from a remittance operator already active in Jalisco. Even a sample of real remittances could help the model validate its synthetic distributions and retrain itself with actual payment behavior as the product builds its own credit history.

6.2 Credit Rate Construction

Although the interest rates developed in this project are based on risk and calibrated using official references, they still rely on some assumptions:

The first, and most notorious factor is the LGD of 45%, which is the parameter most sensitive to the final rate, as demonstrated by the sensitivity analysis. This value was taken as a slightly higher point within the range of secured mortgages (20%-35%) and closer to the range of unsecured consumer credit (close to 84% for the main Mexican banks, according to Moody's Local Mexico, 2025). This was considered due to the partial mitigation provided by collecting payments directly from the remittance receiving account (simulating a normal salary). However, data on the specific recovery rate for this type of product was not found, as it does not yet exist on a large scale.

The financing margin of 300 basis points and the operating cost of 350 basis points were again selected as a midpoint of the range observed for comparable SOFOMEs, according to data from the CNBV (2024) and HR Ratings (2026). For a newly created institution with no track record, these costs would most likely be at the upper end during the first few years of operation.

The 18% ROE target used in the regulatory cost of capital component aligns with the historical range for commercial banks in Mexico and the lower end of the range observed for specialized SOFOMEs (non-bank financial institutions). Given the risk profile of the target segment and the novelty of the product, a more realistic projection for the first few years of operation could use a lower ROE target (14%-16%), which would reduce the cost of capital component and partially offset the higher financing and operating costs described above.

7. Model Limitations & Assumptions & Future Improvements

The principal limitation of this project is the synthetic nature of the dataset. The registry of 10,000 households was constructed by calibrating parametric distributions against official aggregate statistics from Banxico, CONAPO, INEGI, CEMLA, BBVA Research, and CNBV, rather than individual transaction records. Since household-level remittance histories are not publicly available due to confidentiality constraints, the model was trained on simulated behavior designed to be statistically consistent with observed aggregate patterns, but never directly validated against real credit outcomes. This means the true default rate for this specific product remains unknown, as no comparable lending program has operated at scale in Jalisco. Additionally, the model has not been exposed to the full complexity of real-world transaction noise, including irregular transfer schedules, exchange rate volatility, seasonal distortions from individual senders, or idiosyncratic shocks such as illness, job loss, or family disruption on the sender's side.

A second structural assumption concerns the Chow-Lin temporal disaggregation procedure used to convert quarterly municipal remittance data into monthly series. While the aggregation constraint is satisfied with a virtually zero error, and the seasonal correlation between the disaggregated Jalisco series and the national indicator reaches $r = 0.9842$, the monthly values for low-volume municipalities remain model-derived estimates rather than observations. The resulting estimation variance at the municipal level is not captured by the quarterly validation exercise, which may affect the stability of contextual features in Pillar C for smaller municipalities.

Regarding the interest rate construction, the most sensitive parameter is the Loss Given Default (LGD) calibrated at 45%. This figure was set as an intermediate point between secured mortgage recovery ranges (20%–35%) and unsecured consumer credit (close to 84%), justified by the partial mitigation provided by automatic debit collection from the remittance-receiving account. However, no direct empirical evidence exists for the actual recovery rate on this specific product type, since it has not been deployed commercially at scale. Similarly, the funding spread of 300 basis points above the 28-day TIIE and the

operating cost assumption of 350 basis points represent midpoints of ranges observed for comparable SOFOMEs, but a newly created institution with no credit rating would likely face costs at the upper end of those ranges during its initial years of operation.

The financial projection model also assumes a static macroeconomic environment across the three-year horizon, with stable TIIE rates and uninterrupted remittance flows. This is a particularly consequential assumption given that Mexico recorded its first annual decline in remittances since 2013 during 2025 — a 4.6% contraction nationally and nearly 10% in Jalisco — driven by tightened U.S. immigration enforcement and the proposed federal tax on remittances sent abroad (Líder Empresarial, 2026). A sustained reduction in remittance flows would simultaneously erode the income stream underpinning repayment capacity and reduce the quality of the collateral mechanism on which the product's risk structure depends. This systemic risk is the most significant unmodeled assumption in the current projection framework.

Finally, the deployment architecture assumes a fully digital and remote operating model. According to the 2024 National Survey of Financial Inclusion (INEGI-CNBV, 2024), the adoption of digital financial services remains significantly lower in rural areas and among older age groups, both of which are central to the target population. This creates a potential gap between the model's operational assumptions and the on-the-ground realities of customer onboarding and ongoing account management in high-migration municipalities across Jalisco.

Several directions could substantially strengthen the framework in future iterations. The most impactful improvement would be the incorporation of real transactional data from a remittance operator already active in Jalisco. Even a limited sample of actual household-level payment histories would allow the model to validate its synthetic distributions, retrain on genuine behavioral patterns, and produce out-of-sample credit performance metrics that no synthetic calibration can replicate. A second priority would be the implementation of dynamic repricing, linking the product's interest rate to TIIE fluctuations through a variable-rate clause, which would protect the institution's net interest margin in declining rate environments and increase product competitiveness. Third, expanding the geographic scope

beyond Jalisco to other high-migration states Michoacán, Guanajuato, Oaxaca — would require incorporating state-level contextual variables and retraining the regional scoring component, but the modular architecture of the pipeline supports this extension without structural redesign. Finally, integrating behavioral monitoring into the deployed MLflow environment through champion-challenger testing and population stability index (PSI) tracking would allow the institution to detect model drift proactively, ensuring that predictive performance is maintained as the portfolio matures and actual default experience accumulates.

8. Final Discussion & Conclusions

This project set out to answer a concrete question in financial engineering: can the observed pattern of electronic remittance transfers serve as a statistically valid and financially sufficient substitute for credit history in a mortgage-adjacent lending product? The results suggest that it can, under specific conditions, and with meaningful constraints that must be acknowledged alongside the positive findings.

From a modeling perspective, the final selected model *logistic_v2* achieved a ROC-AUC of 0.7872 on the held-out test set, with stable behavior between training and testing partitions and clear probability calibration across the three risk buckets. Logistic Regression was preferred over the ensemble alternatives not because it achieved the highest raw discriminatory power, but because it offered a superior balance of stability, interpretability, and calibration consistency within a regulatory environment where explainability is not optional. The six-component interest rate structure produced risk-adjusted pricing in the range of 28.25% to 47.90% annually, which is substantially below the implicit cost of informal financing available to this segment (100%– 300%+) while remaining above the average APR of conventional mortgage products (14.13%), a result that is financially coherent given the product’s structural differences from traditional secured lending.

From a financial viability perspective, the three-year projection model for the ENR SOFOM demonstrates that the product can generate positive net income from Year 1, recover initial

capital within 1.67 years, and produce an ROE above 18% from Year 2 onward. All 13 financial validation checks are satisfied, including a positive NPV and an equity IRR exceeding the cost of capital. These results position the product as commercially viable, not merely as a social impact initiative. The gross loan portfolio reaches MXN 320.9 million by the end of Year 3, with net income of MXN 29.8 million, demonstrating that financial inclusion and institutional sustainability are not mutually exclusive objectives.

The broader contribution of this work lies in the methodological precedent it establishes. By formalizing remittance flows as alternative collateral through a replicable feature engineering pipeline Chow-Lin disaggregation, three-pillar variable construction, SHAP-guided feature selection, and risk-based pricing integration the project demonstrates that unbanked segments can be evaluated through observable behavioral signals rather than conventional creditworthiness proxies. This is consistent with the frameworks proposed by the World Bank Group (2019) for alternative data in credit scoring, and with the financial inclusion agenda articulated by CONDUSEF (2024) and CEMLA (2025) for remittance-receiving households in Mexico.

Nevertheless, the conclusions must be held with appropriate epistemic caution. The model was trained on synthetic data, and its predictive performance has not been validated against real household-level default outcomes. The macroeconomic context has also shifted materially since the project's baseline was set: the 2025 contraction in remittances to Jalisco driven by U.S. immigration enforcement tightening and the proposed federal tax on international transfers represents a systemic risk that is not captured in the current projection model. Any institution considering operationalizing this framework would need to monitor remittance flow stability as a leading indicator of portfolio health, and design contingency mechanisms for prolonged flow interruptions that go beyond the modeled delinquency scenarios.

In aggregate, this project demonstrates that the financial exclusion of remittance-receiving households in Jalisco is not an inevitable outcome of their economic circumstances, but a consequence of the credit system's failure to recognize the informational content of electronic transfer records. The approximately 4.9 million households nationally and hundreds of

thousands in Jalisco specifically who currently lack access to formal housing credit despite receiving regular, verifiable, and traceable income from abroad represent an addressable market. A machine learning-based credit scoring system calibrated on behavioral remittance signals, combined with risk-based pricing and automatic debit collection, provides a technically grounded and financially sustainable pathway to serving this population. The next step is to move from synthetic validation to real-world piloting, which will require partnership with an active remittance operator and a willing SOFOM willing to originate the first tranche of loans under the proposed framework.

9. References

- Banco de México. (2025). Información de ingresos y egresos por remesas, diciembre de 2024. <https://www.banxico.org.mx/publicaciones-y-prensa/remesas/>
- BBVA. (2025, 14 de noviembre). El impacto social y económico de las remesas en los hogares mexicanos. BBVA México. <https://www.bbva.com/es/mx/economia-y-finanzas/el-impacto-social-y-economico-de-las-remesas-en-los-hogares-mexicanos/>
- BBVA Research. (2025). México. Observatorio Migración y Remesas: cierre anual 2024. https://www.bbvarresearch.com/wp-content/uploads/2025/02/Mexico_Remesas_cierre_anual_2024.pdf
- Centro de Estudios Monetarios Latinoamericanos. (2025). Notas de remesas 06/2025: Concentración de hogares receptores de remesas en pequeñas localidades de México. <https://www.cemla.org/>
- Centro Urbano. (2024, 24 de enero). Infonavit Sin Fronteras, este 2024 paga tu crédito desde EU. <https://centrourbano.com/revista/vivienda/infonavit-sin-fronteras-pagos-eu-2/>
- Cervantes González, J. A., y Ostolaza, R. (2022). ¿Cuántas personas y hogares reciben remesas en México? (Nota de Remesas 06/2022). Centro de Estudios Monetarios Latinoamericanos. <https://www.cemla.org/foroderemesas/notas/2022-06-notas-de-remesas.pdf>
- Comisión Nacional de Vivienda. (2017). Vivienda para mexicanos en el exterior: programa Construye en tu Tierra. Secretaría de Relaciones Exteriores. <https://consulmex.sre.gob.mx/yuma/images/stories/pdfs/ViviendaMigrantes2017.pdf>
- Gramegna, A., y Giudici, P. (2021). SHAP and LIME: An evaluation of discriminative power in credit risk. *Frontiers in Artificial Intelligence*, 4, 752558. <https://doi.org/10.3389/frai.2021.752558>
- Instituto de Información Estadística y Geográfica de Jalisco. (2025, marzo). Ficha informativa: remesas en Jalisco durante el cuarto trimestre de 2024. Gobierno de Jalisco. https://iieg.gob.mx/ns/wp-content/uploads/2025/03/ficha_informativa_remesas_jalisco_T4_2024.pdf
- La Voz de Durango. (2024, 14 de mayo). Más de 12 millones de mexicanos en el extranjero pueden cotizar voluntariamente al Infonavit. <https://lavozdedurango.com.mx/2024/05/14/mas-de-12-millones-de-mexicanos-en-el-extranjero-pueden-cotizar-voluntariamente-al-infonavit/>
- Líder Empresarial. (2026, 4 de febrero). Remesas en Jalisco: ¿Cuánto se percibió en 2025? <https://www.liderempresarial.com/remesas-en-jalisco-cuanto-se-percibio-en-2025/>
- Asociación de Bancos de México. (2018). *Reglas para los requerimientos de capitalización de las instituciones de banca múltiple*. ABM. <https://www.abm.org.mx/legislacion-y-regulacion/juridico/shcp/ReglasparalosrequerimientosdecapitalizaciodelasinstitucionesdeBM.pc>

- Banco de Pagos Internacionales. (2017). *Basilea III: finalización de las reformas poscrisis*. Comité de Supervisión Bancaria de Basilea. https://www.bis.org/bcbs/publ/d424_es.pdf
- Banxico. (2022). *Portal interactivo de información financiera: capital regulatorio bancario*. Banco de México. <https://www.banxico.org.mx/IndicadoresGraficos/actions/contenidoPortal/capital/4/6>
- Banxico. (2025). *Reporte de indicadores básicos de créditos a la vivienda*. Banco de México. <https://www.banxico.org.mx/publicaciones-y-prensa/rib-creditos-a-la-vivienda/>
- Banxico. (2026a). *Tasas de interés representativas: TIE 28 días*. Sistema de Información Económica del Banco de México. <https://www.banxico.org.mx/SieInternet/>
- Chow, G. C., y Lin, A. L. (1971). Best linear unbiased interpolation, distribution, and extrapolation of time series by related series. *The Review of Economics and Statistics*, 53(4), 372–375. <https://doi.org/10.2307/1928739>
- CNBV. (2018). *Basilea III en México*. Comisión Nacional Bancaria y de Valores. [https://www.cnbv.gob.mx/Documents/Basilea%203%20en%20M%C3%A9xico%20\(v16\).pdf](https://www.cnbv.gob.mx/Documents/Basilea%203%20en%20M%C3%A9xico%20(v16).pdf)
- CNBV. (2024). *Boletín estadístico de banca múltiple*. Comisión Nacional Bancaria y de Valores. <https://portafolioinfo.cnbv.gob.mx/>
- CONAPO. (2021). *Índice de Intensidad Migratoria México-Estados Unidos 2020*. Consejo Nacional de Población. <https://www.gob.mx/conapo/documentos/indice-de-intensidad-migratoria-mexico-estados-unidos-2020>
- CONDUSEF. (2018). *Las comisiones bancarias en México*. Comisión Nacional para la Protección y Defensa de los Usuarios de Servicios Financieros. <https://condusef.gob.mx/?idc=379&idcat=1&p=contenido>
- CONDUSEF. (2024). *Calificador de crédito hipotecario*. <https://www.condusef.gob.mx/?p=contenido&idc=1040&idcat=1>
- Diario Oficial de la Federación. (2026, 8 de mayo). *Indicadores económicos: tipo de cambio y tasas de interés interbancarias*. <https://sidof.segob.gob.mx/historicoIndicadores>
- Dinero MX. (2025). *Estadísticas del crédito hipotecario en México 2025*. <https://dinero.mx/credito-hipotecario/estadisticas/>
- El Financiero. (2025, 19 de octubre). *Crédito Mejoravit: qué es y cómo tramitarlo para pagar mejoras y reparación de la casa*. <https://www.elfinanciero.com.mx/mis-finanzas/2025/10/19/credito-mejoravit-que-es-y-como-tramitarlo>
- Hand, D. J., y Henley, W. E. (1997). Statistical classification methods in consumer credit scoring: A review. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 160(3), 523–541. <https://doi.org/10.1111/j.1467-985X.1997.00078.x>
- HR Ratings. (2026). *Banca múltiple en México: análisis sectorial 2025*. HR Ratings de México. https://www.hrratings.com/pdf/SectorialBancos_2025.pdf
- INEGI. (2020). *Censo de Población y Vivienda 2020: principales resultados por localidad (ITER)*. Instituto Nacional de Estadística y Geografía. <https://www.inegi.org.mx/programas/ccpv/2020/>

- Infonavit. (2025). *Plan Estratégico y Financiero 2026*. Instituto del Fondo Nacional de la Vivienda para los Trabajadores. https://portalmx.infonavit.org.mx/wps/wcm/connect/029dbd83-b80b-4616-bf07-6d988f45269d/PEF+2026_VF.pdf
- Investopedia. (2024). *Loss Given Default (LGD): two ways to calculate, plus an example*. <https://www.investopedia.com/terms/l/lossgivendefault.asp>
- Kagan, J., y Khartit, K. (2020, 1 de diciembre). Risk-based pricing: what it means, how it works. *Investopedia*. <https://www.investopedia.com/terms/r/riskbased-pricing.asp>
- Moody's Local México. (2025, 9 de octubre). *BBVA México, S.A.: revisión anual de calificación*. Moody's Local México. https://moodylocal.com.mx/wp-content/uploads/2025/10/1.2.1_MLMX_Informe-BBVA-102025.pdf
- Mosley, L., Eckley, I., y Gibberd, A. (2022). Sparse temporal disaggregation. *Statistical Methods & Applications*, 31(3), 831–860. <https://doi.org/10.1007/s10260-021-00604-3>
- Office of the Comptroller of the Currency. (2007). *Loss given default of high loan-to-value residential mortgages* (Working paper 2007-4). U.S. Department of the Treasury. <https://www.comptrollerofthecurrency.gov/publications-and-resources/publications/economics/working-papers-archived/pub-econ-working-paper-2007-4.pdf>
- Sofomes.com. (2025). *Guía completa SOFOM México 2026*. <https://sofomes.com/>
- Stiglitz, J. E., y Weiss, A. (1981). Credit rationing in markets with imperfect information. *The American Economic Review*, 71(3), 393–410. <https://www.jstor.org/stable/1802787>
- Thomas, L. C. (2000). A survey of credit and behavioural scoring: Forecasting financial risk of lending to consumers. *International Journal of Forecasting*, 16(2), 149–172. [https://doi.org/10.1016/S0169-2070\(00\)00034-0](https://doi.org/10.1016/S0169-2070(00)00034-0)
- Tu Hipoteca Fácil. (2021). *¿Qué es una Sofom y cómo te ayuda a comprar casa?* <https://www.tuhipotecafacil.com/post/que-es-una-sofom-y-como-te-ayuda-a-comprar-casa>
- Vera-Jaramillo, A. (2025). *Temporal disaggregation methods in practice: an applied review*. INEGI Working Papers Series.
- Wall Street Prep. (2023). *Loss Given Default (LGD): formula and calculator*. <https://www.wallstreetprep.com/knowledge/loss-given-default-lgd/>
- World Bank Group. (2019). *Credit scoring approaches guidelines*. World Bank Publications. <https://thedocs.worldbank.org/en/doc/935891585869698451-0130022020/original/CREDITSSCORINGAPPROACHESGUIDELINESFINALWEB.pdf>
- Banco de México. (2024). *Indicadores básicos de créditos personales y microcréditos: datos a febrero de 2024*. Banco de México. <https://www.banxico.org.mx/publicaciones-y-prensa/rib-creditos-personales/{01469D83-2BB3-A5CB-95E4-6B11ABEE2DED}.pdf>
- BBVA México. (s.f.). *Tipos de crédito hipotecario en México*. BBVA México. <https://www.bbva.mx/educacion-financiera/creditos/credito-hipotecario/tipos-de-credito-hipotecario.html>

- CEMLA. (2025). *Notas de remesas 06/2025: concentración de hogares receptores de remesas en pequeñas localidades de México*. Centro de Estudios Monetarios Latinoamericanos. <https://www.cemla.org/>
- Dinero MX. (2025). *Estadísticas del crédito hipotecario en México 2025*. <https://dinero.mx/credito-hipotecario/estadisticas/>
- INEGI y CNBV. (2024). *Encuesta Nacional de Inclusión Financiera (ENIF) 2024*. Instituto Nacional de Estadística y Geografía y Comisión Nacional Bancaria y de Valores. <https://www.inegi.org.mx/rnm/index.php/catalog/1081>
- SEDATU. (2024). *Programa Nacional de Vivienda 2021-2024*. Secretaría de Desarrollo Agrario, Territorial y Urbano. <https://www.gob.mx/sedatu/documentos/programa-nacional-de-vivienda-2021-2024>

10. Appendices

The complete source code, deployment configuration, machine learning pipeline, and web application files developed for this project are available in the following GitHub repository:

[ppmelch/Final_Credit_Model: Final Credit Model project](https://github.com/ppmelch/Final_Credit_Model)